MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

An athletic coach told his team that muscle multiplied with speed equal to power. What dimensions does he view for muscle?

(A) MLT^{-2}

(B) $\mathrm{ML}^{2}\,\mathrm{T}^{-2}$

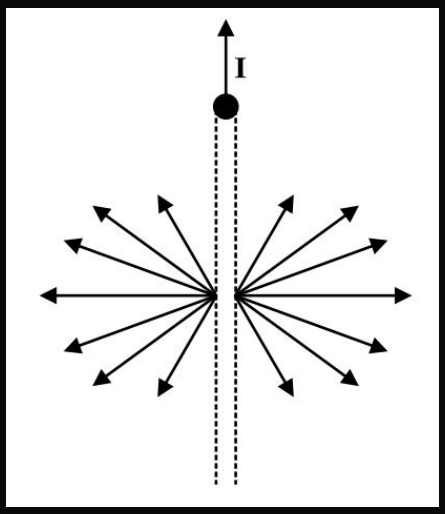
(C) MLT^{-3}

(D) L

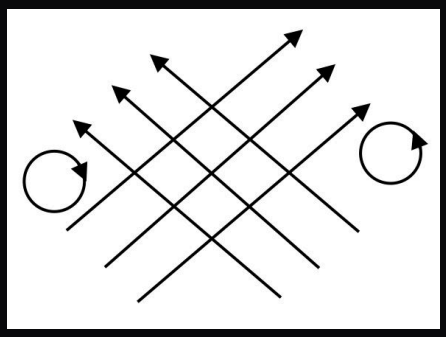
Q.2

Point out the correct direction of the magnetic field in the given figures.

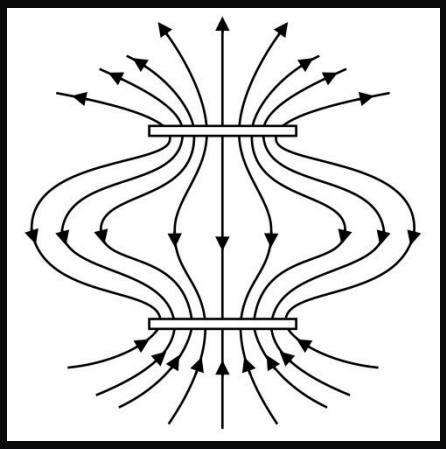
(A)



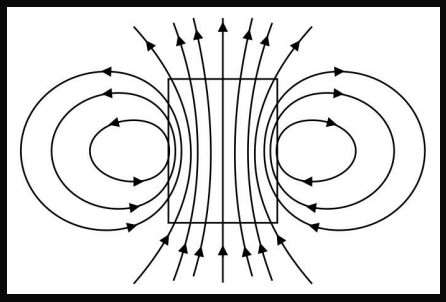
(B)



(C)



(D)



Q.3

A clock face has negative charges $-q, -2q, -3q, \ldots, -12q$ fixed at the position of the corresponding numerals on the dial. The clock hands do not disturb the net field due to point charges. At what time does the hour hand point in the same direction of the electric field at the centre of the dial?

(A) $3:30$

(B) $9:30$

(C) $6:30$

(D) $12:30$

Q.4

The energy that should be added to an electron to reduce its de-Broglie wavelength from $1\ \text{nm}$ to $0.5\ \text{nm}$ is

(A) four times the initial energy.

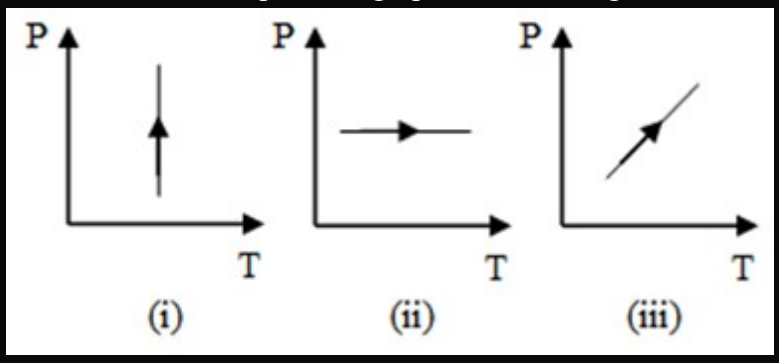
(B) equal to the initial energy.

(C) twice the initial energy.

(D) thrice the initial energy.

Q.5

Pressure versus temperature graphs of an ideal gas are as shown in figure. Choose the wrong statement



(A) density of gas is increasing in graph (i)

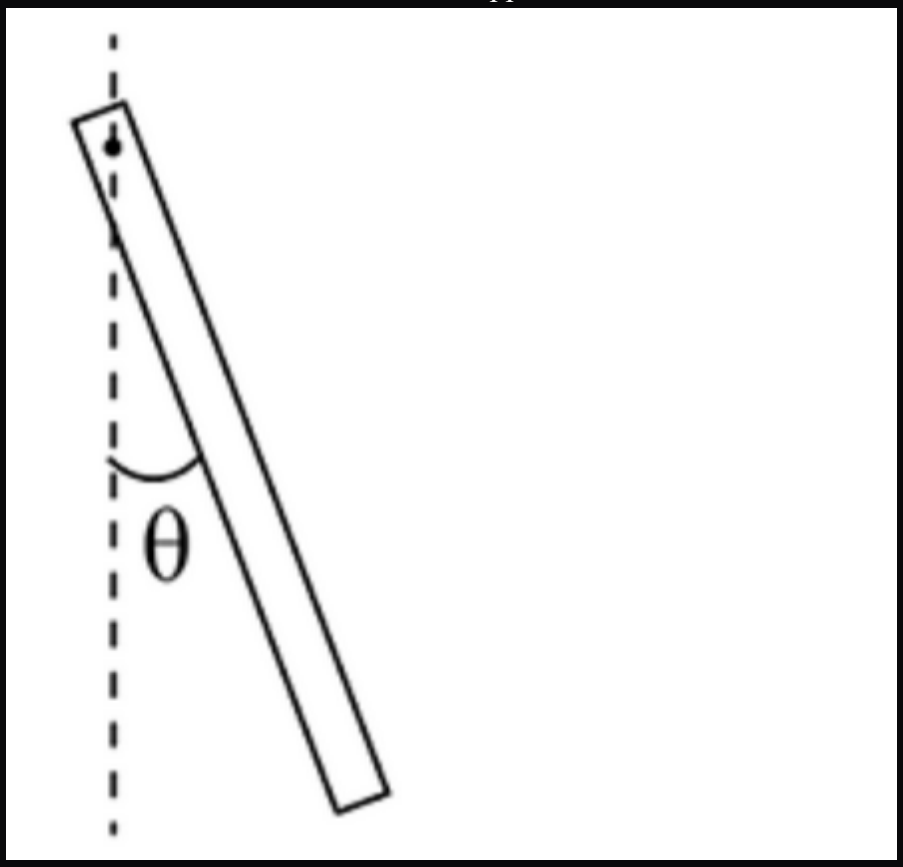
(B) density of gas is decreasing in graph (ii)

(C) density of gas is constant in graph (iii)

(D) none of these

Q.6

A thin rod of mass m hinged at its upper end is to be maintained in equilibrium, making an angle θ with the vertical. The minimum force that needs to be applied to its bottom end to make this possible is :



(A) $\frac{mg}{2\sin\theta}$

(B) $\frac{mg}{\cos\theta}$

(C) $\frac{mg\sin\theta}{2}$

(D) $mg\sin\theta$

Q.7

The heat generated in a circuit is given by $Q = I^2\ Rt$, where I is current, R is resistance, and t is time. If the percentage errors in measuring I, R, and t are $2\%,\ 1\%$ and 1% respectively, then the maximum error in measuring heat will be

(A) 2%

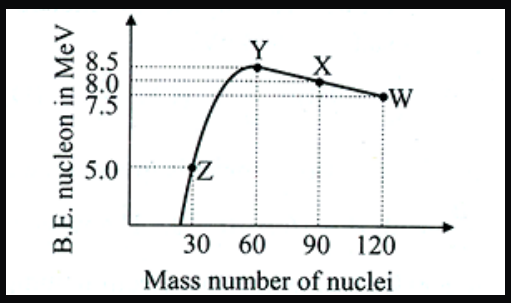
(B) 3%

(C) 4%

(D) 6%

Q.8

Binding energy per nucleon v/s mass number curve for nuclei is shown in the figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is



(A) $Y \rightarrow 2Z$

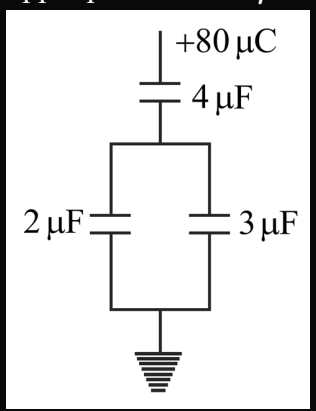
(B) $W \rightarrow X + Z$

(C) $W \rightarrow 2Y$

(D) $X \rightarrow Y + Z$

Q.9

In the given circuit, a charge of $+80\ \mu C$ is given to upper plate of a $4\mu F$ capacitor. At steady state the charge on the upper plate of the $3\mu F$ capacitor is



(A) $60\ \mu C$

(B) $48\ \mu C$

(C) $80\ \mu C$

(D) $0\ \mu C$

Q.10

Assertion: Although both, a stone and a feather, are dropped from the same elevation, they do not reach the ground at the same time.
Reason: Acceleration due to gravity, depending upon the mass of the object, varies from object to object.

(A) If both assertion and reason are true and reason is the correct explanation of assertion.

(B) If both assertion and reason are true but reason is not the correct explanation of assertion.

(C) If assertion is true but reason is false.

(D) If both assertion and reason are false.

Q.11

Eight equal drops of water are falling through air with a steady velocity of 10 cms^{-1}. If the drops combine to form a single drop big size, then the terminal velocity of this big drop is

(A) $80\ \text{cms}^{-1}$

(B) $30\ \text{cms}^{-1}$

(C) $10\ \text{cms}^{-1}$

(D) $40\ \text{cms}^{-1}$

Q.12

A refracting telescope consists of

(A) one concave mirror and one convex lens.

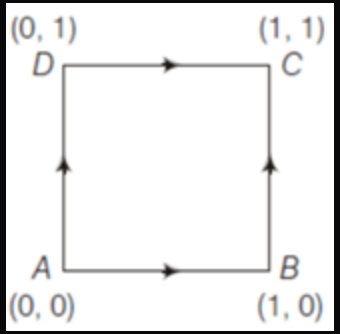
(B) two convex lenses of equal focal length.

(C) two concave mirrors of different focal lengths.

(D) two convex lenses of unequal focal lengths.

Q.13

A particle is moving in a force field given by $\mathbf{F} = y^2\hat{\mathbf{i}} - x^2\hat{\mathbf{j}}$. Starting from A, the particle has to reach C either along ABC or ADC. Let the workdone along the two paths be W_1 and W_2, respectively. Then, (W_1, W_2) are



(A) $(-1, 1)$

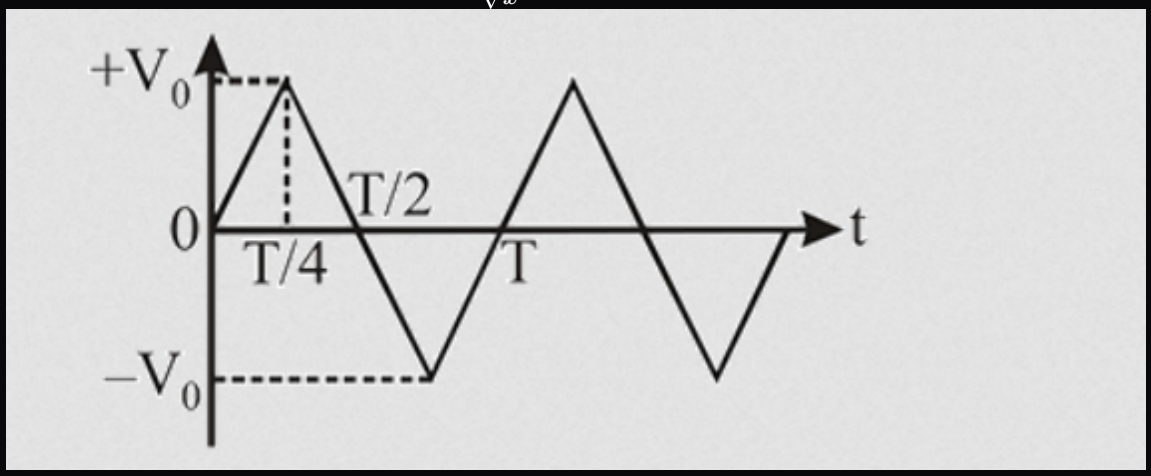
(B) $(-1, 0)$

(C) $(1, 1)$

(D) $(-1, -1)$

Q.14

The voltage time (V-t) graph for triangular wave having peak value V_0 is as shown in figure. The rms value of V in time interval from $t = 0$ to $T/4$ is $\frac{V_0}{\sqrt{x}}$ then find the value of x.



(A) 5

(B) 4

(C) 7

(D) 3

Q.15

In a photodiode, the conductivity increases when the material is exposed to light. It is found that conductivity changes only if the wavelength is less than 600 nm. What is the bandgap?

(A) 2.06 eV

(B) 1.2 eV

(C) 3.4 eV

(D) 4.2 eV

Q.16

Consider the following statements

I. If a piece of bar magnet is broken into two equally long pieces along length, then the pieces will not lose the magnetic properties.

II. Magnetic properties of a substance lie at the atomic level.

Which of the statement(s) given above is/are correct?

(A) Only I

(B) Only II

(C) Both I and II

(D) Neither I nor II

Q.17

Suppose a container is evacuated to leave just one molecule of gas in it. Let, v_{a} and v_{rms} represent the average speed and the RMS speed of the gas.

(A) $v_{\mathrm{a}} > v_{\mathrm{rms}}$

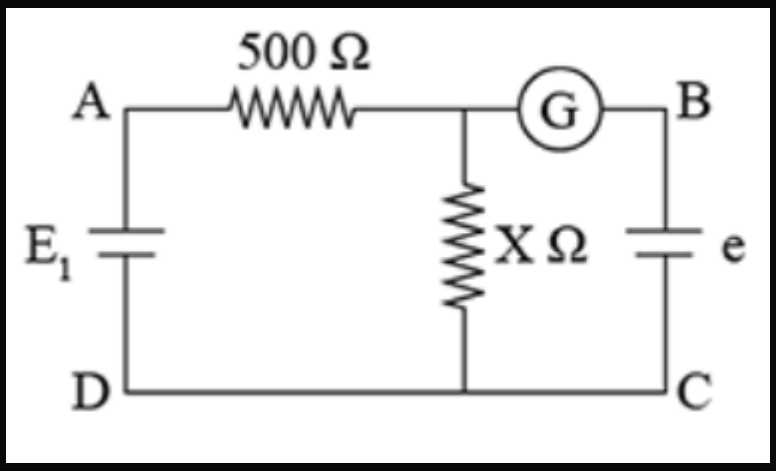
(B) $v_{\mathrm{a}} < v_{\mathrm{rms}}$

(C) $v_{\mathrm{a}} = v_{\mathrm{rms}}$

(D) v_{rms} is undefined.

Q.18

In an experiment according to set up, when $\mathrm{E}_1 = 12\ \mathrm{V}$ and internal resistance zero, $\mathrm{e} = 2\ \mathrm{V}$. The galvanometer reads zero, then X would be-



(A) 200

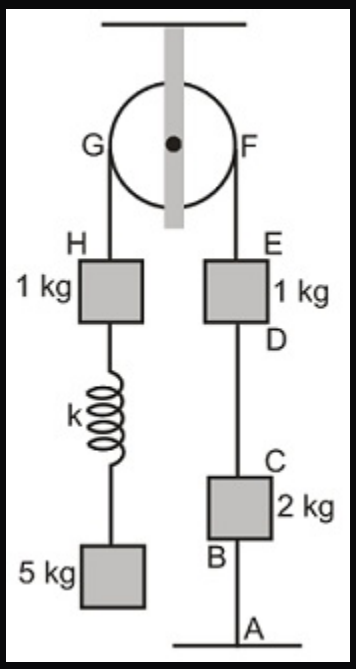
(B) 500

(C) 100

(D) 10

Q.19

In the situation shown in the diagram, the system is in equilibrium. All the strings, spring and pulley are light. Just after cutting the string *AB*,



(A) The tension in the string GH is $6\ g$

(B) The tension in the string GH is zero

(C) The tension in string GH is $\frac{21g}{4}$

(D) The $2\ \text{kg}$ block remains in equilibrium

Q.20

A rigid body is in pure rotation, that is undergoing fixed axis rotation. Then which of the following statements are true -

(A) You can find two points in the body in a plane perpendicular to the axis of rotation having same velocity

(B) You can find two points in the body in a plane perpendicular to the axis of rotation having same acceleration

(C) Speed of all the particles lying on the curved surface of a cylinder whose axis coincides with the axis of rotation is same

(D) none of these

Q.21

Four resistance of $3\ \Omega,\ 3\ \Omega,\ 3\ \Omega$ and $4\ \Omega$ respectively are used to form a Wheatstone bridge. The $4\ \Omega$ resistance is short-circuited with a resistance R in order to get bridge balanced. The value of R will be

Q.22

A double-slit arrangement produces interference fringes that are 0.25° apart using wavelength $600\ \text{nm}$. If the angular separation is increased to 0.3° by changing the wavelength of the light used, then the new wavelength must be (Mark answer in nm)

Q.23

Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A emits 10^4

times the power emitted from B. The ratio (λ_A/λ_B) of their wavelengths λ_A and λ_B at which the peaks occur in their respective radiation curves is

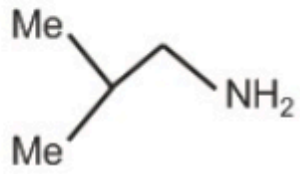
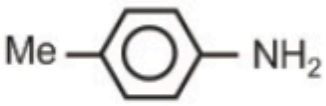
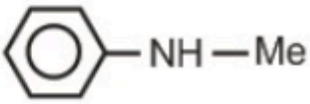
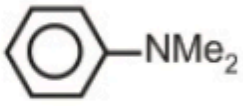
Q.24

If on transition to the ground state an He^+ ion emits two photons in succession, having wavelengths $1026.7\ \mathring{A}$ and $304\ \mathring{A}$, then the quantum number n corresponding to the exciting state of He^+ ion is [$R = 1.096 \times 10^7\ m^{-1}$]

Q.25

In the Coolidge tube experiment, if the applied voltage is increased to three times, the short wavelength limit of continuous $X-$ ray spectrum shifts by 20 pm. What is the initial voltage (in kV) applied to the tube? (mark answer to nearest integer)

Q.26

	Column I		Column II
S. NO.	Compound		Characteristics
a.		p.	Liebermann's nitroso reaction
b.		q.	Evolution of N_2 with HNO_2
c.		r.	Dye test
d.		s.	Green colour with HNO_2
		t.	Carbylamine test

(A) (a $\longrightarrow$ q, t; b $\longrightarrow$r, t; c $\longrightarrow$ p; d $\longrightarrow$ s)

(B) (q $\longrightarrow$ q, r; c $\longrightarrow$t, a; b $\longrightarrow$ q; a $\longrightarrow$ r)

(C) (a $\longrightarrow$ b, a; b $\longrightarrow$r, c; t $\longrightarrow$ s; r $\longrightarrow$ c)

(D) (a $\longrightarrow$ c, q; a $\longrightarrow$t, b; r $\longrightarrow$ a; c $\longrightarrow$ t)

Q.27

The boiling point of an azeotropic mixture of water and ethyl alcohol is less than that of theoretical value of water and alcohol mixture. Hence, the mixture shows

(A) that solution is highly saturated

(B) positive deviation from Raoult's law

(C) negative deviation from Raoult's law

(D) nothing can be said

Q.28

The first $(\Delta_{i}H_{1})$ and second $(\Delta_{i}H_{2})$ ionization enthalpies (in $kJmol^{-1}$) and the electron gain enthalpy $(\Delta_{eg}H)$ (in $kJmol^{-1}$) of the elements I, II, III, IV and V are given below

Element	$\Delta_{i}H_{1}$	$\Delta_{i}H_{2}$	$\Delta_{eg}H$
1	520	7300	-60
II	419	3051	-48
III	1681	3374	-328
IV	1008	1846	-295
V	2372	5251	$+48$

the least reactive non- metal and the most reactive metal of these are respectively

(A) IV and V

(B) II and V

(C) V and III

(D) V and II

Q.29

The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, $K^{+}, Ca^{2+}Mg^{2+}, Be^{2+}$?

(A) $Ca^{2+} < Mg^{2+} < Be^{+} < K^{+}$

(B) $Mg^{2+} < Be^{2+} < K^{+} < Ca^{2+}$

(C) $Be^{2+} < K^{+} < Ca^{2+} < Mg^{2+}$

(D) $K^{+} < Ca^{2+} < Mg^{2+} < Be^{2+}$

Q.30

The average atomic mass of a mixture containing 79 mole % of ^{24}Mg remaining 21 mole % of ^{25}Mg and ^{26}Mg is 24.31% . Mole of ^{26}Mg is -

(A) 5

(B) 20

(C) 10

(D) 15

Q.31

Which of the following is an example of aldol condensation?

(A) $2CH_3COCH_3 \xrightarrow{\text{Dil.NaOH}} CH_3COHCH_3CH_2COCH_3$

(B) $2HCHO \xrightarrow{\text{Dil.NaOH}} CH_3OH$

(C) $C_6H_5CHO + HCHO \xrightarrow{\text{Dil.NaOH}} C_6H_5CH_2OH$

(D) None of the above

Q.32

Choose the one molecule or ion from each set having smallest bond angle.
(i) NH_3, PH_3 or AsH_3
(ii) O_3^+, O_3
(iii) NO_2^- or O_3
(iv) $X - S - X$ angle in $SOCl_2$ or SOF_2

(A) $NH_3, O_3^+ O_3, SOCl_2$

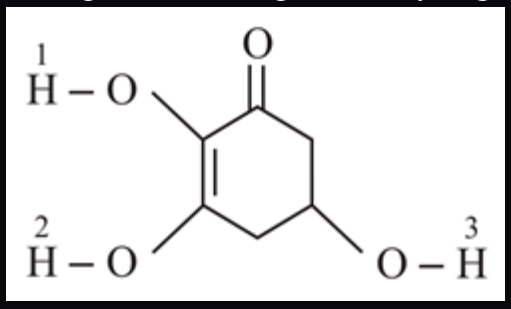
(B) $AsH_3, O_3, NO_2^-, SOF_2$

(C) PH_3, O_3^+, NO_2^-

(D) AsH_3, O_3^+, O_3, SOF_2

Q.33

Arrange the following marked hydrogen according to decreasing order of acidic strengths.



(A) $2 > 1 > 3$

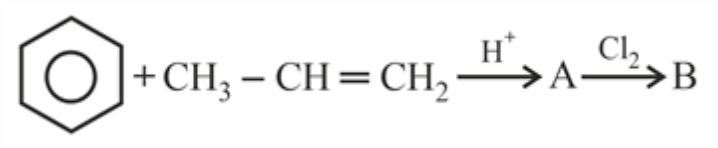
(B) $1 > 2 > 3$

(C) $1 > 3 > 2$

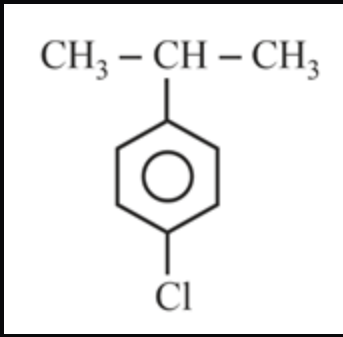
(D) $2 > 3 > 1$

Q.34

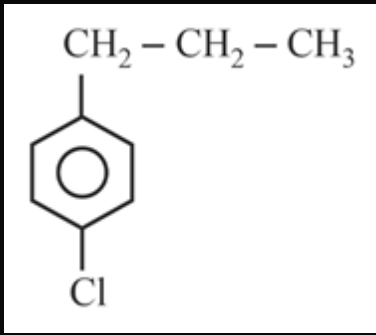
What is the product (major) B in the following reaction?



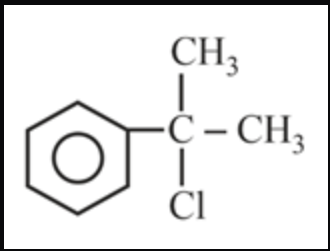
(A)



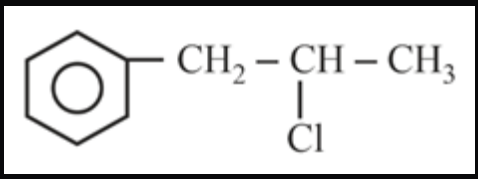
(B)



(C)



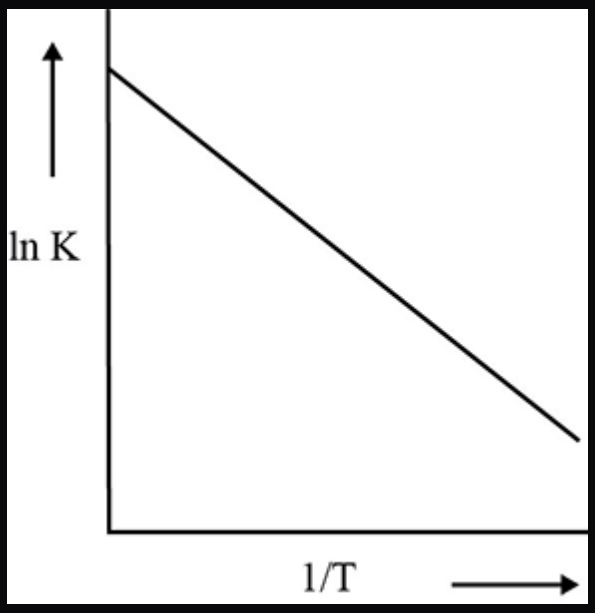
(D)



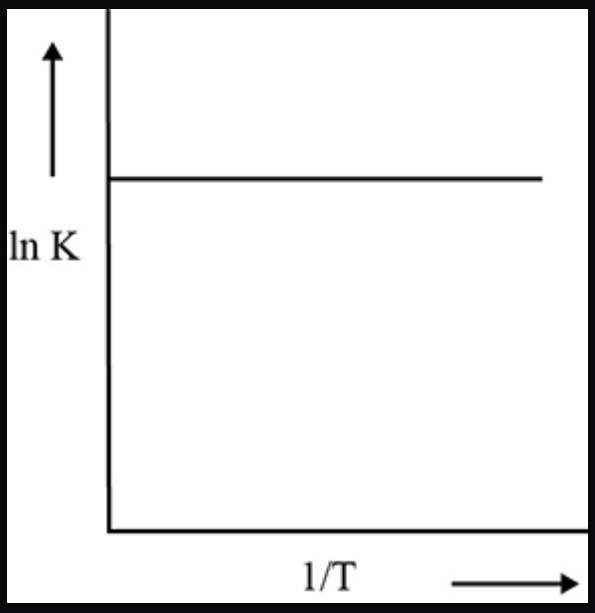
Q.35

Which In K vs $1/T$ plot is correct for an equilibrium that shifts toward reactants at higher temperatures?

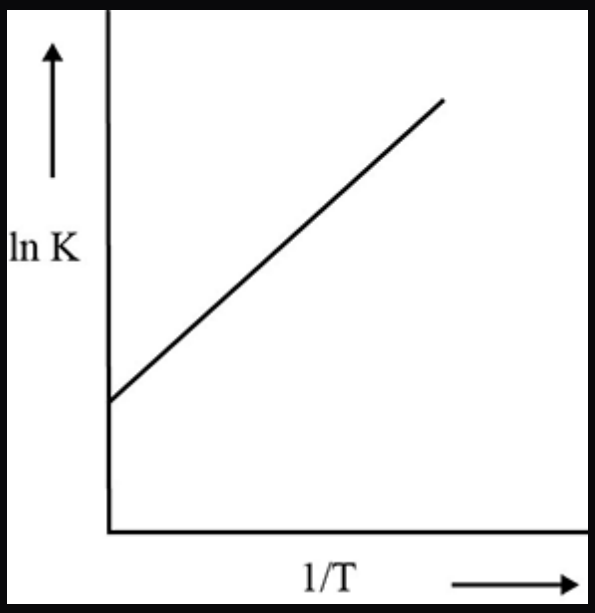
(A)



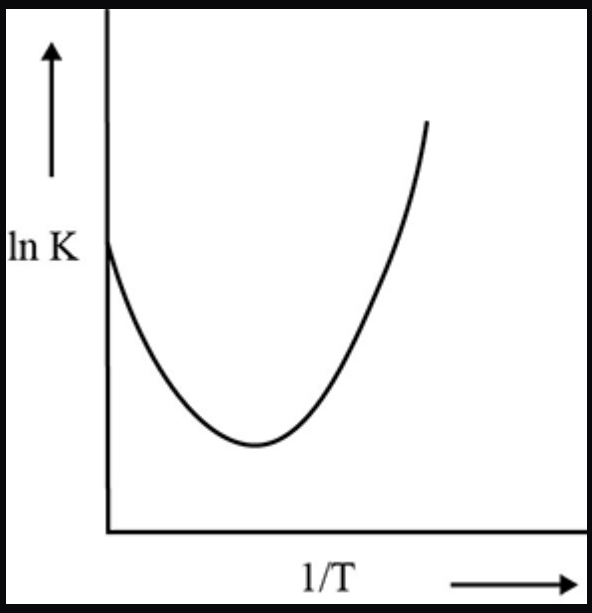
(B)



(C)



(D)



Q.36

The oxidation states of Cr in $[\mathrm{Cr}\,(\mathrm{H_2O})_6]\,\mathrm{Cl_3}$, $[\mathrm{Cr}\,(\mathrm{C_6H_6})_2]$ and $\mathrm{K_2}\,[\mathrm{Cr(CN)_2(O)_2}\,(\mathrm{O_2})\,(\mathrm{NH_3})]$ respectively are

(A) $+3, +4$ and $+6$

(B) $+3, +2$ and $+4$

(C) $+3, 0$ and $+6$

(D) $+3, 0$ and $+4$

Q.37

The qualitative distinction of $\mathrm{ZnSO_4}$ and $\mathrm{Al_2}\,(\mathrm{SO_4})_3$ can be done by using the reagent :

(A) $\mathrm{NH_4OH}$

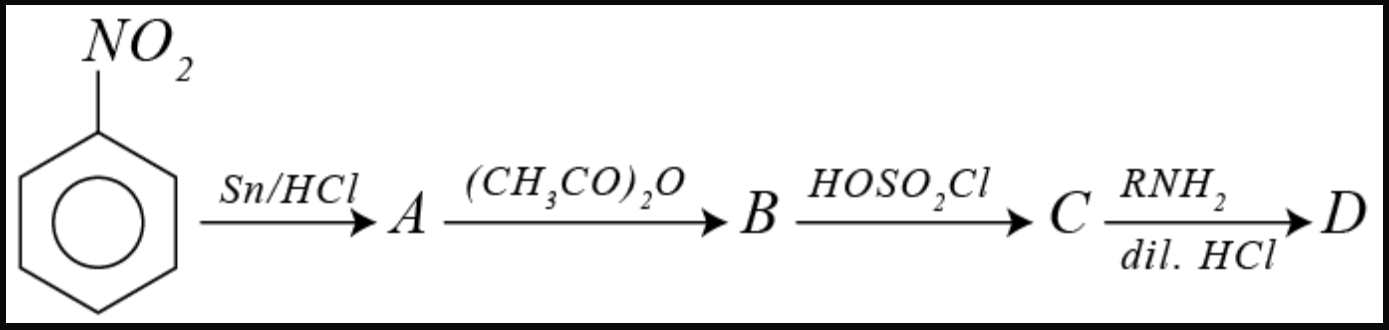
(B) NaOH

(C) Any of these

(D) none of these

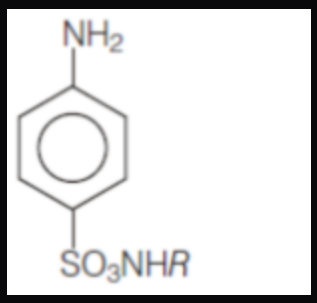
Q.38

Consider the following sequence of reactions:

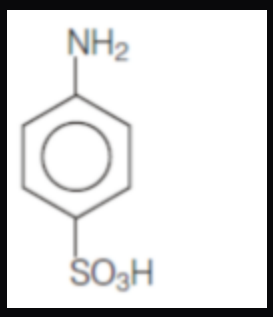


The correct structure of products (D) is

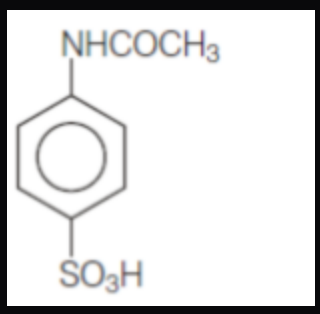
(A)



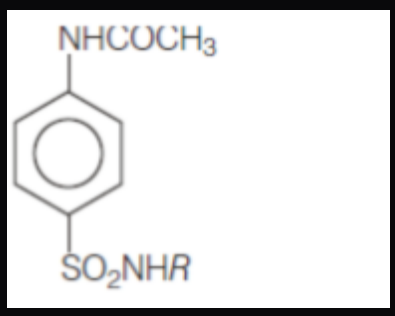
(B)



(C)



(D)



Q.39

STATEMENT-1: $NO_2 + CO \rightarrow CO_2 + NO$

$r = k[NO_2]$

The rate of the above reaction is independent of the concentration of CO

STATEMENT-2: The rate does not depend upon [CO] because it is involved in fast step.

(A) STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is correct explanation of STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is NOT correct explanation of STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.40

Which of the following is not correctly matched?

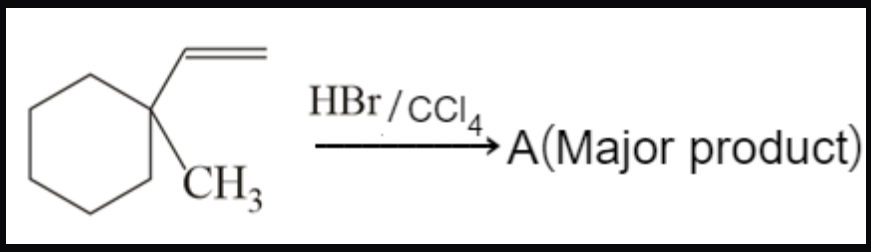
(A) Acidic oxides P_2O_5, NO_2, Cl_2O_7

(B) Basic oxides Na_2O, CaO, MgO

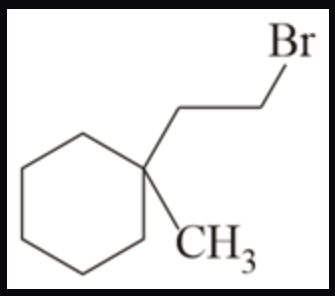
(C) Neutral oxides CO_2, CO, BeO

(D) Amphoteric oxides ZnO, SnO, Al_2O_3

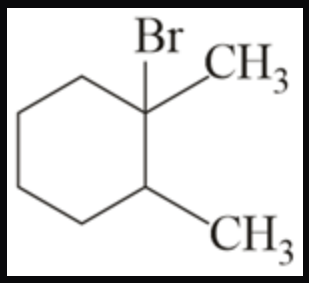
Q.41



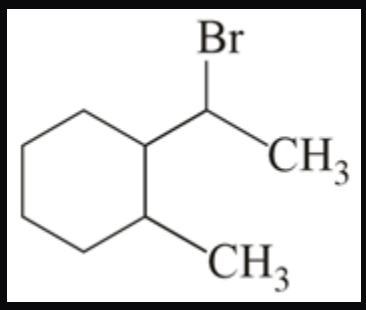
(A)



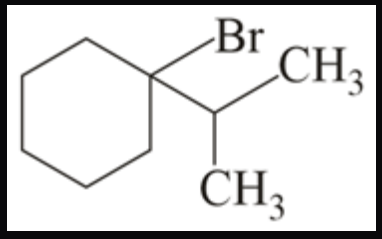
(B)



(C)



(D)



Q.42

Pick out incorrect statement

(A) In a electrolysis experiment, α -amino acids migrate at the isoelectric point towards electrodes

(B) p-aminobenzenesulphonic acid as a dipolar ion; while p-aminobenzoic acid does not

(C) Sulphanilic acid is soluble in base, but not in acid

(D) H_3NCH_2COOH is more acidic than RCH_2COOH

Q.43

The number of π electrons present in 6.4 g of calcium carbide is $-$ (N_A = Avagadro's number)

(A) $4\ N_A$

(B) $0.4\ N_A$

(C) $0.1\ N_A$

(D) $0.2\ N_A$

Q.44

The addition of $NaOH$ to Cr^{3+} solution produces the precipitate of

(A) $Cr(OH)_3$

(B) CrO_3

(C) $Cr_2O_3\ (H_2O)_n$

(D) Cr_2O_3

Q.45

Which of the following is the predominant intermolecular force of attraction in HF?

(A) Dipole-induced dipole interaction

(B) Ion-dipole interaction

(C) Dipole-dipole interaction

(D) London forces

Q.46

The freezing point of a solution, prepared from $1.25gm$ of a non-electrolyte and $20gm$ of water, is $271.9\ K$. If molar depression constant is $1.86\ K$ mole $^{-1}$, then molar mass of the solute will be (Round off to the nearest integer)

Q.47

The number of spectral lines that can be possible when electrons in 7th shell in different hydrogen atoms return to the 2nd shell, is

Q.48

A reaction is found to be second order w.r.t. one of the reactants & has rate constant of $0.5\ \mathrm{mol}^{-1}\mathrm{dm}^3\ \mathrm{min}^{-1}$. If initial concentration is $0.2\ \mathrm{moldm}^{-3}$ then $\mathrm{t}_{1/2}$ of reaction is (in min)

Q.49

The total number of cyclic structural as well as stereo isomers possible for a compound with the molecular formula $\mathrm{C_5H_{10}}$ is

Q.50

The EAN of metal atoms in $\left[\mathrm{Fe(CO)_2\ (NO^+)_2}\right]$ is 'x' and EAN of metal atoms in $\mathrm{Co_2(CO)_8}$ is 'y'. The value of $\mathrm{x + y}$ is

Q.51

There are 3 identical black balls, 4 identical white balls and 2 identical red balls. The number of ways they can be arranged in a row so that at least one ball separates the balls of the same colour is

(A) 1210

(B) 1212

(C) 1244

(D) 1254

Q.52

Let $H : \frac{(5x-y-24)^2}{26} - (x+5y-10)^2 = 26$ is a hyperbola, then

(A) length of transverse axis is $\sqrt{26}$.

(B) length of conjugate axis is 1.

(C) center is $(1, 5)$

(D) one vertex is $(0, 2)$

Q.53

If $f(x) = \cos x - \int_0^x (x-t) f(t) dt$, then $f''(x) + f(x)$ is equal to

(A) $-\cos x$

(B) $-\sin x$

(C) $\int_0^x (x-t) f(t) dt$

(D) 0

Q.54

The minimum value of the function, $f(x)=x^{3/2}+x^{-3/2}-4\left(x+\frac{1}{x}\right)$. for all permissible real values of x, is

(A) -10

(B) -6

(C) -7

(D) -8

Q.55

The least integral value of K for which $(K-2)x^2+8x+k+4>\sin^{-1}(\sin 12)+\cos^{-1}(\cos 12)\forall x\in R$, is

(A) -7

(B) -5

(C) -3

(D) 5

Q.56

Three ordinary and fair dice are rolled simultaneously. The probability of the sum of outcomes being atleast equal to 8, is equal to

(A) $\frac{81}{108}$

(B) $\frac{27}{216}$

(C) $\frac{81}{216}$

(D) $\frac{181}{216}$

Q.57

Let $\bar{a}=\bar{i}+\bar{j}, \bar{b}=2\bar{i}-\bar{k}$. Then the vector $\bar{r}$ satisfying the equations $\bar{r}\times\bar{a}=\bar{b}\times\bar{a}$ and $\bar{r}\times\bar{b}=\bar{a}\times\bar{b}$ is

(A) $\bar{i}-\bar{j}+\bar{k}$

(B) $3\bar{i}-\bar{j}+\bar{k}$

(C) $3\bar{i}+\bar{j}-\bar{k}$

(D) $\bar{i}-\bar{j}-\bar{k}$

Q.58

The number of solution of the equation $a^{f(x)}+g(x)=0$, where $a>0, g(x)\neq 0$ and has minimum value $\frac{1}{2}$, is -

(A) Infinitely many

(B) Only one

(C) Two

(D) Zero

Q.59

$\lim_{x \to 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4\left(e^{2x^4} - 1 - 2x^4\right)}$ is equal to

(A) 0

(B) $-\frac{1}{6}$

(C) $\frac{1}{6}$

(D) does not exist

Q.60

The mean of the three numbers is 15. The range of this data is 8 while the difference between the two smallest numbers is 1. The greatest of the three numbers is:

(A) 22

(B) 20

(C) 18

(D) 24

Q.61

If the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{1} = \frac{y-6}{2} = \frac{z+\lambda}{2}$ intersect, then value of λ is

(A) -9

(B) 9

(C) 8

(D) -8

Q.62

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, a, b, \in N$. Then :

(A) there exist more than one but finite number of B's such that $AB = BA$

(B) there exists exactly one B such that $AB = BA$

(C) there exists infinitely many B 's such that $AB = BA$

(D) there cannot exist any B such that $AB = BA$ C

Q.63

If $\frac{xdy}{dx} - y = \sqrt{x^2 + y^2}$, then

(A) $x + \sqrt{x^2 + y^2} = cy^2$

(B) $\sqrt{x^2+y^2}-y=cx^2$

(C) $\sqrt{x^2+y^2}+y=cx^2$

(D) $\sqrt{x^2+y^2}-x=cy^2$

Q.64

Let $f(x)=\begin{cases}\max\{|x|,x^2\} & |x|\le 3\\ 12-|x| & 3<|x|\le 12\end{cases}$. If S is the set of points in the interval $(-12,12)$ at which f is not differentiable, then S is

(A) equal to $\{-3,3\}$

(B) equal to $\{-3,-1,1,3\}$

(C) an empty set

(D) equal to $\{-3,-1,0,1,3\}$

Q.65

There are 100 sets named $A_1, A_2, \ldots, A_{100}$ such that $n(A_i)=i+2$ represents the number of elements in A_i and $A_1 \subset A_2 \subset A_3 \ldots \subset \ldots \subset A_{100}$. If $\bigcap_{i=3}^{100} A_i = A$ then $n(A)=$

(A) 3

(B) 4

(C) 5

(D) 6

Q.66

If 6th term in the expansion of $\left(\frac{3}{2}+\frac{x}{3}\right)^n$ is numerically greatest, when $x=3$, then find the sum of possible integral values of n.

(A) 23

(B) 24

(C) 25

(D) 26

Q.67

The area between the curve given in parametric form: $x=a(\theta-\sin\theta), y=a(1-\cos\theta), 0\le\theta<2\pi$ and the x-axis is

(A) πa^2

(B) $2\pi a^2$

(C) $3\pi a^2$

(D) $4\pi a^2$

Q.68

The radius of the circle $\operatorname{Re}\left(\frac{iz+1}{iz-1}\right)=2$ is

(A) 1

(B) $\sqrt{2}$

(C) 2

(D) $\frac{1}{\sqrt{2}}$

Q.69

One card is missing from a normal pack of 52 playing cards. A card is drawn from the remaining pack of cards. If a spade card is obtained, then the chance of getting a spade card by drawing another card from remaining 50 cards is

(A) $\frac{1}{17}$

(B) $\frac{2}{17}$

(C) $\frac{3}{17}$

(D) $\frac{4}{17}$

Q.70

A circle touches the lines $y=\frac{x}{\sqrt{3}}, y=x\sqrt{3}$ and has unit radius. If the centre of this circle lies in the first quadrant, then one possible equation of this circle is -

(A) $x^2+y^2-2x(\sqrt{3}+1)-2y(\sqrt{3}+1)+8+4\sqrt{3}=0$

(B) $x^2+y^2-2x(\sqrt{3}+1)-2y(\sqrt{3}+1)+5+4\sqrt{3}=0$

(C) $x^2+y^2-2x(\sqrt{3}+1)-2y(\sqrt{3}+1)+7+4\sqrt{3}=0$

(D) $x^2+y^2-2x(\sqrt{3}+1)-2y(\sqrt{3}+1)+6+4\sqrt{3}=0$

Q.71

If p is a positive integer and f be a function defined for positive numbers and attains only positive values such that $f(xf(y))=x^py^4$ then find p.

Q.72

If $f\left(x+y\right)=f\left(x\right)+3y^2+kxy,\ \ f\left(1\right)=1, f\left(2\right)=6$ and $f(0)\neq -2$, then the value of $\int\limits_{-1}^{2} f\left(x\right)dx$ is

Q.73

If $\mathrm{A}=\begin{bmatrix} 1 & 25 \end{bmatrix}, \mathrm{B}=\begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}, \mathrm{C}=\begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, \mathrm{D}=\begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}, \mathrm{E}=$

$\begin{bmatrix} x^2-5x+20 \\ x+2 \end{bmatrix}$ and $\mathrm{F}=[40]$ are matrices and α, β are roots of equation $AB^5C^{10}D^5E=F$ then the value of $(1-\alpha)(1-\beta)$ is

Q.74

The number of real roots of the equation $x^{2013}+e^{2014x}=0$ is

Q.75

Let $a_1, a_2, a_3 \ldots a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1}=416$ and $a_9+a_{43}=66$ if $a_1^2+a_2^2+\ldots\ldots+a_{17}^2=140\ m$, then m is equal to $\left(\text{given that } 1^2+2^2+\ldots\ldots+n^2=n(n+1)(2n+1)/6\right)$

Answers & Solutions

Q.1 Answer:

—

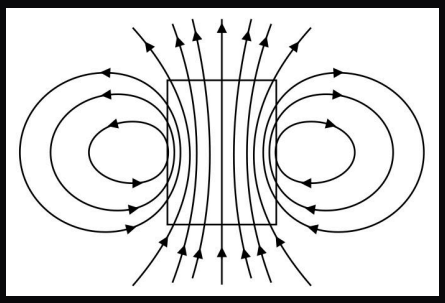
Solution:

Accordind to the question

$\text{Muscle} \times \text{Speed} = \text{Power}$

$\text{Muscle} = \frac{\text{Power}}{\text{Speed}} = \text{MLT}^{-2}$

Q.2 Answer:



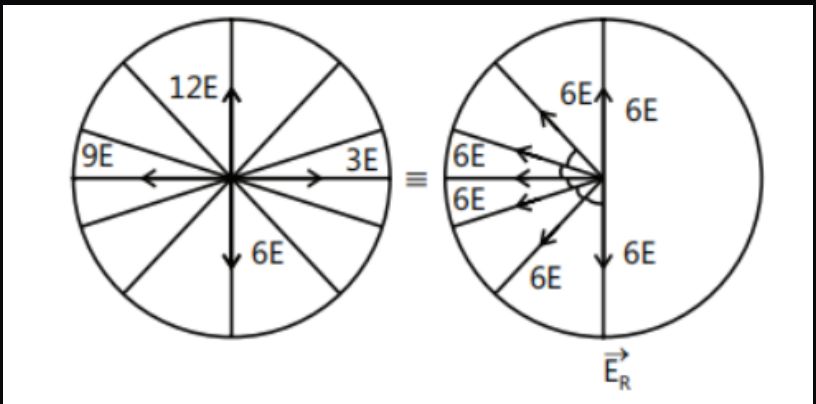
Solution:

The magnetic field lines never cross each other ,and they form closed loops. They travel from north pole to south pole outside the magnet and from south pole to north pole inside the magnet. Hence, the last option is the correct one.

Q.3 Answer:

9 : 30

Solution:



Electric field due to various charges is shown in the diagram.

So, on solving, net electric field will be along the line as shown in the given diagram.

The charges at $8:00, 10:00$ **will be at angle** 30° **below and above the horizontal line and at** $7:00, 11:00$ **will be at angle** 60° **below and above the horizontal line.**

On resolving the components along x **&** $y-$**axes,**

$\tan\theta = \frac{P}{B}$

$\tan\theta = \frac{6E}{6E+2\times6E\cos(30^\circ)+2\times6E\cos(60^\circ)}$

$\tan\theta = \frac{6E}{6E+6E+6E\sqrt{3}}$

$\tan\theta = \frac{1}{2+\sqrt{3}}$

$\tan\theta = 2-\sqrt{3}$

Hence, $\theta = 15^\circ$**.**

So, the net electric field will be above $9:00$ **at angle** 15°**.**

The angle between two consecutive numbers in a clock is 30°**.**

So, the direction will be in the middle of $9:00$ **&** $10:00$**.**

Hence, at $9:30$**.**

Q.4 Answer:

thrice the initial energy.

Solution:

de-Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\therefore \quad \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}} \Rightarrow \frac{1\times10^{-9}}{0.5\times10^{-9}} = \sqrt{\frac{E_2}{E_1}}$$

$$\Rightarrow 2 = \sqrt{\frac{E_2}{E_1}} \Rightarrow \frac{E_2}{E_1} = 4$$

$$\therefore E_2 = 4E_1$$

$\therefore$ **Energy to be added** $= E_2 - E_1$

$= 4E_1 - E_1 = 3E_1$

Q.5 Answer:

none of these

Solution:

(i) At constant temperature $\mathrm{P} \propto \rho$

(ii) At constant pressure $\mathrm{T} \propto \frac{1}{\rho}$

(iii) $\mathrm{V} \rightarrow$ **constant**

$\therefore \rho \rightarrow$ **constant**

Q.6 Answer:

$\frac{mg \sin \theta}{2}$

Solution:

Minimum force is required when this force is applied perpendicular to the rod balancing torque about the hinge $mg\frac{l}{2}\sin\theta = Fl$

$\Rightarrow \quad F = \frac{mg\sin\theta}{2}$

Q.7 Answer:

6%

Solution:

$\mathrm{Q} = \mathrm{I}^2\,\mathrm{Rt}$

Required error is $2 \times 2\% + 1\% + 1\% = 6\%$

Q.8 Answer:

$\mathrm{W} \rightarrow 2\mathrm{Y}$

Solution:

The total binding energy of Y**,**

$E_{\mathrm{Y}} = 60 \times 8.5 = 510\ \mathrm{MeV}$

The total binding energy of W**,**

$E_{\mathrm{W}} = 120 \times 7.5 = 900\ \mathrm{MeV}$

So in the process, $\mathrm{W} \rightarrow 2\mathrm{Y}$**,**

$2E_{\mathrm{Y}} > E_{\mathrm{W}}$

So, the energy will be released in this process.

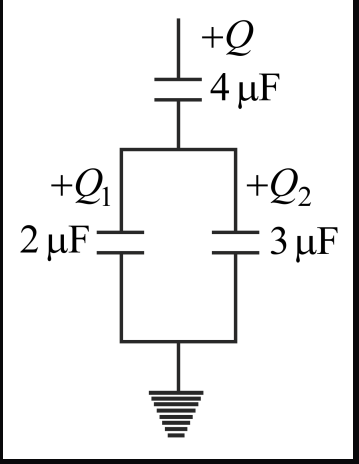
Q.9 Answer:

$48\ \mu\mathrm{C}$

Solution:

Given, charge on the upper plate of $4\ \mu\mathrm{F}$ **capacitor,** $Q = +80\ \mu\mathrm{C}$**.**

Let Q_1 **be the charge on** $2\ \mu\mathrm{F}$ **and** Q_2 **be the charge on** $3\ \mu\mathrm{F}$ **capacitors.**



By Junction law,

$Q = Q_1 + Q_2 \quad \ldots . (1)$

Potential difference on $2\ \mu$F and $3\ \mu$F are equal because they are connected in parallel.

$\therefore \frac{Q_1}{2} = \frac{Q_2}{3} \Rightarrow Q_1 = \frac{2Q_2}{3}$

By putting this value in (1), we get

$Q = \frac{2Q_2}{3} + Q_2 = \frac{5Q_2}{3} \Rightarrow Q_2 = \frac{3Q}{5} = \frac{3\times 80}{5} \Rightarrow Q_2 = 48\ \mu\text{C}$

$\therefore$ The charge on the upper plate of $3\ \mu$F is $48\ \mu$C.

Note: A student can solve this problem using charge divider circuit: $Q_2 = \frac{C_2}{C_1+C_2}Q$

Q.10 Answer:

If assertion is true but reason is false.

Solution:

Acceleration due to gravity is independent of the mass of the object. The stone and feather dropped from the same height do not reach the ground at the same time because of air resistance.

Q.11 Answer:

$40\ \text{cms}^{-1}$

Solution:

Terminal velocity is proportional to square of radius of the sphere.

$v \propto r^2$

$\frac{v_1}{v_2} = \frac{r_1^2}{r_2^2}$

and using volume conservation,

$8 \times \frac{4}{3}\pi r_1^3 = \frac{4}{3}\pi r_2^3 \Rightarrow r_2 = 2r_1$

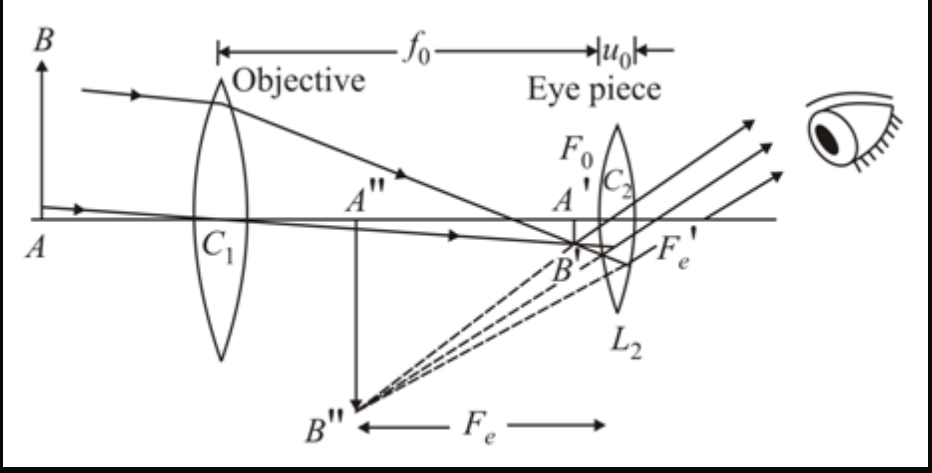
therefore, $\text{v}_2 = 4\text{v}_1 = 40\ \text{cms}^{-1}$

Q.12 Answer:

two convex lenses of unequal focal lengths.

Solution:

A refracting telescope (also called a refractor) is a type of optical telescope that uses a two convex lens to form an image.



Lens near the object is known as objective lens and lens near our eyes are known as eyepiece lens. Objective lens is bigger than eyepiece lens so that clear image is obtained.

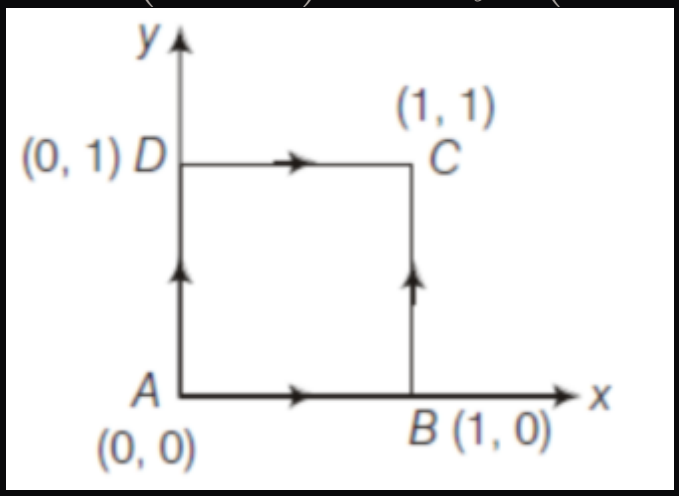
Q.13 Answer:

$(-1, 1)$

Solution:

$W_{ABC} = W_{AB} + W_{BC}$

$= \int_{x=0}^{x=1} \left(y^{2\hat{i}} - x^2\hat{j}\right) \cdot dx\hat{i} + \int_{y=0}^{y=1} \left(y^2\hat{i} - x^2\hat{j}\right) \cdot dy\hat{j}$



For $AB, y = 0$ **and** $BC, x = 1$

$W_{ABC} = \int_{X=0}^{X=1} \left(0\hat{i} - x^2\hat{j}\right) \cdot dx\hat{i} + \int_{y=0}^{y=1} \left(y^2\hat{i} - \hat{j}\right) \cdot dx\hat{j}$

$= 0 + (-1)\int_0^1 dy = -1$

Now, $W_{ADC} = W_{AD} + W_{DC}$

$= \int_{y=0}^{y=1} \left(y^{2\hat{i}} - x^2 j\right) \cdot dy\hat{j} + \int_{y=0}^{x=1} \left(y^2\hat{i} - x^2\hat{j}\right) \cdot dx\hat{i}$

For $AD, x = 0$ **and for** $DC, y = 1$

$\Rightarrow W_{ADC} = \int_0^1 \left(y^{2\hat{i}} - 0\right) dy\hat{j} + \int_0^1 \left(1\hat{i} - x^{2\hat{j}}\right) dx\hat{i}$

$= 0 + \int_0^1 dx = 1$

Hence, $(W_1, W_2) = (-1, 1)$

Q.14 Answer:

3

Solution:

$V = \frac{V_0}{T/4} t \Rightarrow V = \frac{4\,V_0}{T} t$

$\Rightarrow V_{rms} = \sqrt{< V^2 \rangle} = \frac{4\,V_0}{T} \sqrt{< t^2 >} = \frac{4\,V_0}{T} \left\{ \frac{\int_0^{T/4} t^2 dt}{\int_0^{T/4} dt} \right\} = \frac{V_0}{\sqrt{3}}$

Q.15 Answer:

$2.06\ \text{eV}$

Solution:

Given,

The maximum wavelength of light, which increases the conductivity, $\lambda = 600\ \text{nm}$

Hence the energy of this photon must be equal to band gap (E_g**).**

$\therefore\ E_g = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{600 \times 10^{-9}}\ \text{J}$

$= \frac{6.6 \times 3 \times 10^{-17}}{600 \times 1.6 \times 10^{-19}}\ \text{eV} = 2.06\ \text{eV}$

Q.16 Answer:

Only II

Solution:

A magnetic monopole does not exist, i.e., even if we break a magnet it will behave as two magnets, and we cannot isolate a magnetic monopole.

If we keep on dividing the magnet till the atomic level the magnetic property will still be there, thus the magnetic property of a substance lie in the atomic level.

Q.17 Answer:

$v_a = v_{rms}$

Solution:

After taking the mean of velocities, we get mean velocity.

The root-mean-square speed is the measure of the speed of particles in a gas, defined as the square root of the mean velocity-squared of the molecules in a gas.

Then if,

$u_a = \frac{u_1 + u_2 + \ldots}{n}$

$u_{rms} = \sqrt{\frac{u^2{}_1 + u^2{}_2 + \ldots.}{n}}$

For one molecule,

$u_a = \frac{u_1}{1}\ \text{and}\ u_{rms} = \sqrt{\frac{u^2{}_1}{1}}$

$\Rightarrow u_a = u_{rms}$

Q.18 Answer:

100

Solution:

Voltage across X is $2v$

So $\left(\frac{E_1}{500+X}\right) X = 2$

$$\Rightarrow \left(\frac{12}{500+X}\right) X = 2$$

$$\Rightarrow 12X = 1000 + 2X$$

$$\Rightarrow 10X = 1000$$

$$X = 100\Omega$$

Q.19 Answer:

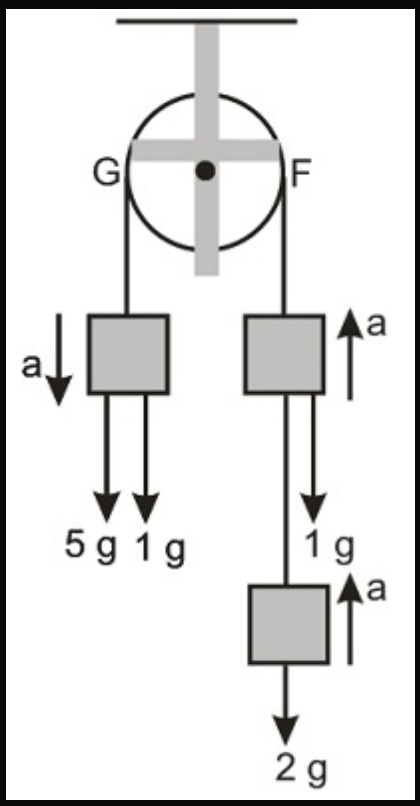
The tension in string GH is $\frac{21g}{4}$

Solution:

In equilibrium, just before the string AB is cut the tension in spring is $5g$.

Just after the string AB is cut the spring force won't change but all blocks except of 5 kg start accelerating.

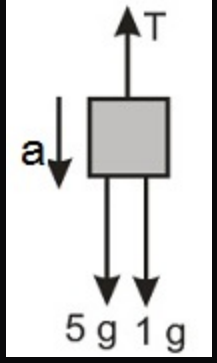
Rough free body diagram is shown below



$$a = \frac{\text{Net pulling force}}{\text{Total mass}}$$

$$= \frac{5g+g-g-2g}{4} = \frac{3g}{4}$$

For 1 kg block (connected to spring)



Free body diagram would be as shown

$5g + g - T = 1 \times \frac{3g}{4}$

$T = 6g - \frac{3g}{4} = \frac{21g}{4}$

Q.20 Answer:

Speed of all the particles lying on the curved surface of a cylinder whose axis coincides with the axis of rotation is same

Solution:

All points in the body, in plane perpendicular to the axis of rotation revolve in connection circles. All points lying on circle of same radius have same speed (and also same magnitude of acceleration) but different directions of velocity (also different directions of acceleration). Hence there cannot be two points in the given plane with same velocity or with same acceleration. As mentioned above, points lying on circle of same radius have same speed.

Angular speed of body at any instant w.r.t. any point on body is same by definition.

Q.21 Answer:

—

Solution:

The bridge will be balanced when the shunted resistance of the value of $3\ \Omega$

$\therefore 3 = \frac{4 \times \mathrm{R}}{4+\mathrm{R}} \Rightarrow 12 + 3\mathrm{R} = 4\mathrm{R} \Rightarrow R = 12\ \Omega$

Q.22 Answer:

—

Solution:

Angular fringe width is given by, $\theta = \frac{\lambda}{d}$**.**

If the slit width remains constant, angular fringe width (θ) **is directly proportional to the wavelength** (λ) **of the light which is used.**

Therefore, we can write,

$\frac{\theta_1}{\theta_2} = \frac{\lambda_1}{\lambda_2}$

$\Rightarrow \frac{\theta_1}{\theta_2} = \frac{0.25}{0.3} = \frac{600}{\lambda_2}$

$\Rightarrow \lambda_2 = \frac{600 \times 0.3}{0.25} = 720\ \mathrm{nm}$

Q.23 Answer:

—

Solution:

$\left(\frac{dQ}{dt}\right)_A = 10^4 \left(\frac{dQ}{dt}\right)_B$
$(400R)^2 T_A^4 = 10^4 \left(R^2 T_B^4\right)$
So, $2T_A = T_B$ **and** $\frac{\lambda_A}{\lambda_B} = \frac{T_B}{T_A} = 2$

Q.24 Answer:

—

Solution:

$\frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = RchZ^2\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$
Put $\lambda_1 = 1026.7\text{Å}$ **and** $\lambda_2 = 304$ Å
$Z = 2$ **for** He^+ **ion**
On solving for n
$n = 6$

Q.25 Answer:

—

Solution:

$\because$ **cut-off wavelength for continuous x-ray is given as :** $\lambda_{\min} = \frac{hc}{eV_0}$
$\therefore \frac{\mathrm{hc}}{\mathrm{eV}_0} = \lambda \;\&\; \frac{\mathrm{hc}}{3\mathrm{eV}_0} = \lambda - \Delta\lambda$
$(\mathrm{V}_0 = \text{accelerating voltage})$

$\Rightarrow \quad \mathrm{V}_0 = \frac{2\mathrm{hc}}{3\mathrm{e}\Delta\lambda} = 41\ \mathrm{kV}$

Q.26 Answer:

(a $\rightarrow$ q, t; b $\rightarrow$r, t; c $\rightarrow$ p; d $\rightarrow$ s)

Solution:

(a $\rightarrow$ q, t) 1° aliphatic amine with HNO_2 gives alcohol and N_2 and also gives carbylamine test.
(b $\rightarrow$ r, t) 1° aromatic amine gives dye test and also gives carbylamine test.
(c $\rightarrow$ p) 2° aromatic gives N - nitroso compound with HNO_2, which gives Liebermann's nitroso reaction.
(d $\rightarrow$ s) 3° aromatic gives p - nitroso compound which is green in colour.

Q.27 Answer:

positive deviation from Raoult's law

Solution:

The boiling point of an azeotropic mixture of water and ethyl alcohol is less than that of theoretical value of water and alcohol mixture. Hence, the mixture shows positive deviation from Raoult's law.
Positive deviations from Raoult's law are noticed when
(i) Exp. vaue of vapour pressure of mixture is more than calculated value.

(ii) Exp. value of boiling point of mixture is less than calculated value.
(iii) $\Delta H_{mixing} = +ve$
(iv) $\Delta V_{mixing} = +ve$

Q.28 Answer:

V and II

Solution:

Generally metals having low I.E.are more reactive and $\Delta_{eg}H = +$ **ve for noble gas.**

Q.29 Answer:

$K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$

Solution:

Smaller the size and higher the charge more will be polarising power of cation. Since the order of the size of cation is $K^+ > Ca^{2+} > Mg^{2+} > Be^{2+}$

So the correct order of polarising power is :

$K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$

Q.30 Answer:

10

Solution:

Let $\%$ **mole of** ^{26}Mg **be** X

$\frac{(21-X)25+26x+79\times24}{100} = 24.31$

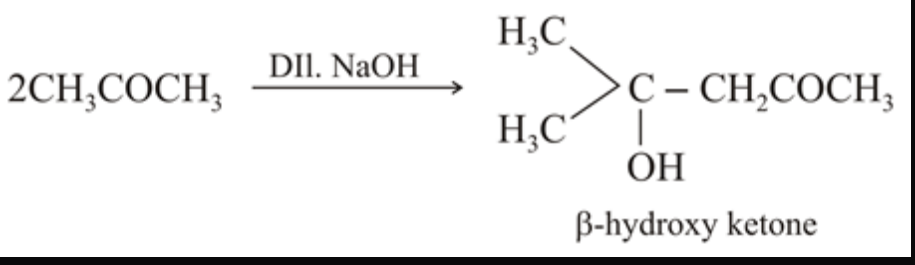
$X = 10\ \%$

Q.31 Answer:

$2CH_3COCH_3 \xrightarrow{Dil.NaOH} CH_3COHCH_3CH_2COCH_3$

Solution:

Aldehyde and ketone having α**-hydrogen atom undergo aldol condensation in presence of dilute base to give** $\beta-$**hydroxy aldehydes or ketones. Aldol condensation reaction results in the formation of a carbon-carbon bond between enol(or enolate) molecule with aldehyde or ketone. Acetone enolate is the nucleophile and another molecule of acetone is electrophile in this reaction.**
The addition step without subsequent dehydration is an aldol reaction.
Acetone has α**-hydrogen atom, hence it will give aldol condensation reaction.**



This β-hydroxy ketone further undergoes condensation reaction to form α, β unsaturated ketone.

Q.32 Answer:

$AsH_3, O_3, NO_2^-, SOF_2$

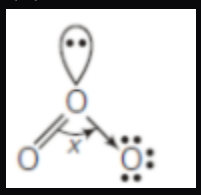
Solution:

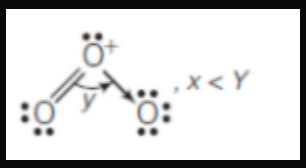
(i) According to Drago rule, bond angle of NH_3, PH_3 and AsH_3 decreases top to bottom

$NH_3 > PH_3 > AsH_3$

Here, AsH_3 has smallest bond angle.

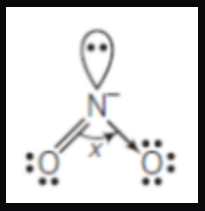
(ii)

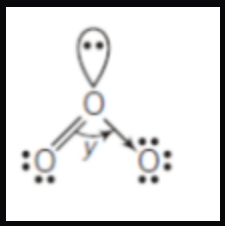




Lone pair-double bond repulsion in O_3 is greater than single electrondouble bond repulsion in O_3^+. Therefore, the repulsion between lone-pair single bond in O_3 is greater than repulsion between single electron single bond in O_3

(iii) According to Bent rule, $N-O$ bond in NO_2^- has more p-character than $O-O$ bond in O_3.

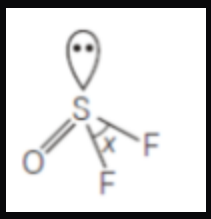


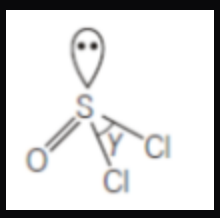


O_3 has sp^2 - hybridisation on central oxygen atom.

(iv) According to Bent rule, p-character of $S-Cl$ bond in $SOCl_2$ is lower than p-character of $S-F$

bond in SOF_2.



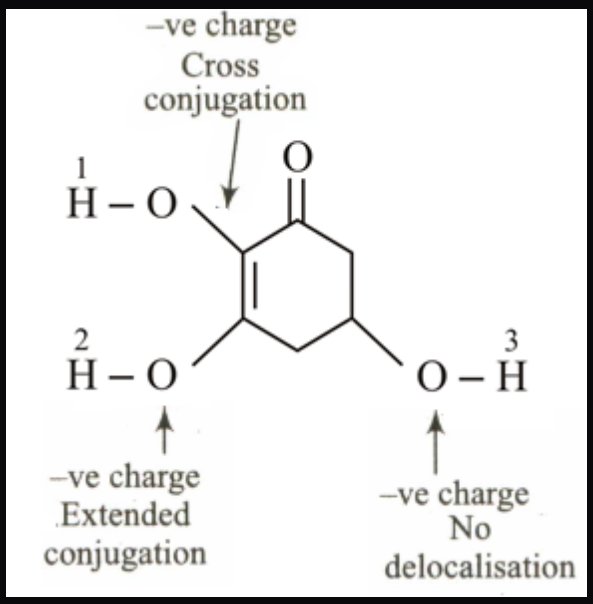


Hence, the correct choice regarding smallest bond angle is (b).

Q.33 Answer:

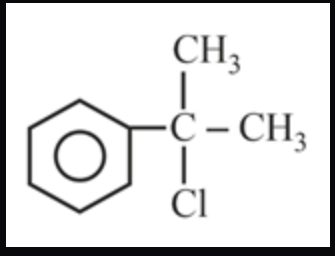
$2 > 1 > 3$

Solution:



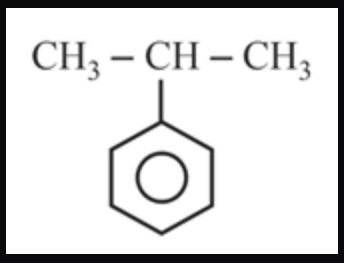
The negative charge formed on the 1st oxygen atom is involving in the cross conjugation, the negative charge formed on the 2nd oxygen atom is involving in extended conjugation and the negative charge formed at 3rd oxygen atom in not involving in the resonance. Due to the cross conjugation the stability of anion is less in comparison with extended conjugation. More the stability of negative charge more is the acidic strength of the compound. Hence, the order of acidic strength is 2>1>3

Q.34 Answer:

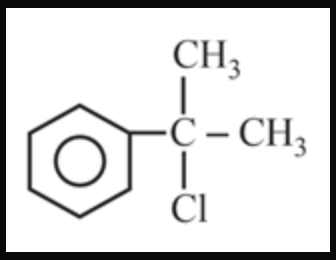


Solution:

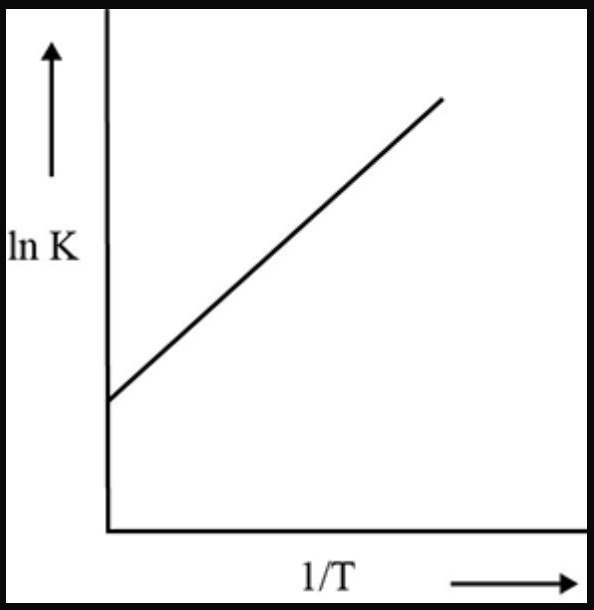
A is isopropyl benzene (Cumene)



When photo chlorination of cumene is carried out then chlorine radical get substituted to that carbon where free radical is most stable (major product) therefore, structure of B is



Q.35 Answer:



Solution:

$\mathrm{InK} = -\frac{\Delta \mathrm{H}^{\circ}}{\mathrm{R}}\left(\frac{1}{\mathrm{T}}\right) + \frac{\Delta \mathrm{S}^{\circ}}{\mathrm{R}}$

At high temperature if shifted backward, it is exothermic reaction.

$\Delta \mathrm{H}^{\circ}_{\mathrm{reac}}$ **is -ve**

Slope is +ve

Q.36 Answer:

$+3, 0$ and $+6$

Solution:

$[\mathrm{Cr}\left(\mathrm{H_2O}\right)_6]\,\mathrm{Cl_3} : \mathrm{x} + 6 \times 0 + 3 \times -1 = 0$

$\mathrm{x} = +3$

$[\mathrm{Cr}\left(\mathrm{C_6H_6}\right)_2] : \mathrm{x} + 2 \times 0$

$\mathrm{x} = 0$

$\mathrm{K_2}\left[\mathrm{Cr(CN)_2(O)_2}\left(\mathrm{O_2}\right)\left(\mathrm{NH_3}\right)\right]$

$2 \times (+1) + x + 2 \times (-1) + 2 \times (-2) + 2 \times (-1) + 0 = 0$

$2 + x - 2 - 4 - 2 = 0$

$x = +6$

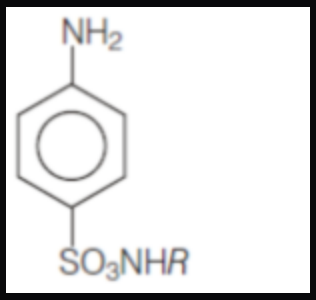
Q.37 Answer:

NH_4OH

Solution:

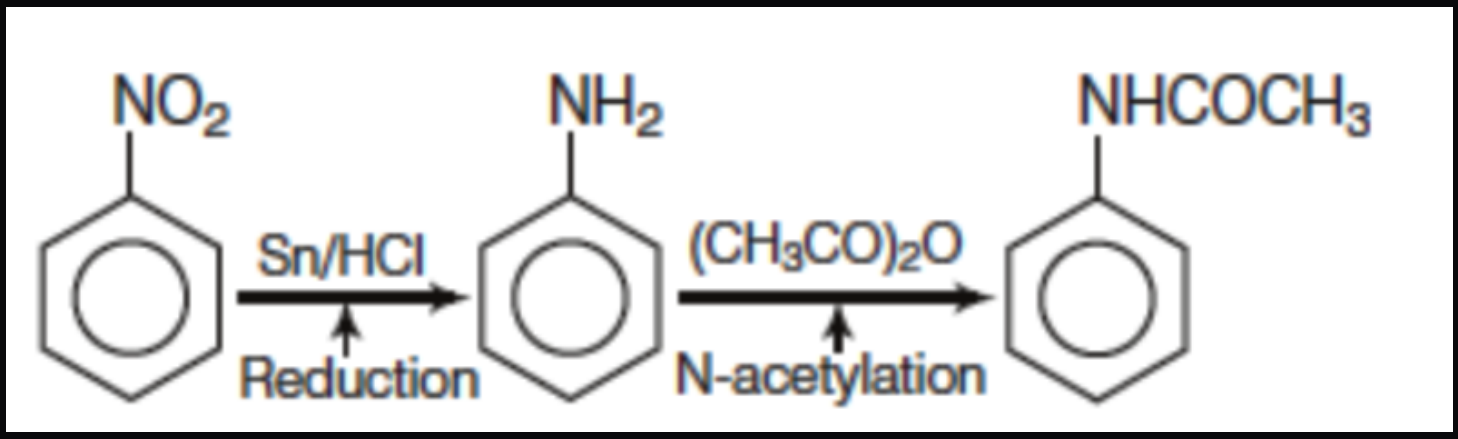
Qualitative distinction of $ZnSO_4$ and $Al_2(SO_4)_3$ is done using $NH_4OH + NH_4Cl$.

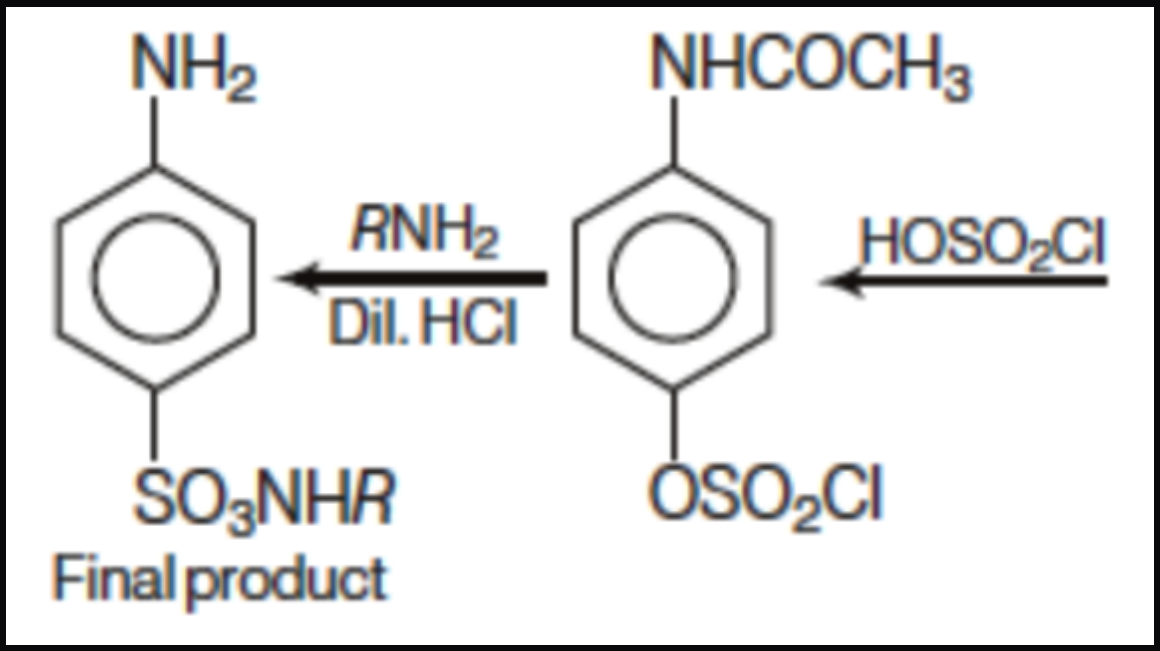
Q.38 Answer:



Solution:

The reaction sequence involves various processes are shown as:





Q.39 Answer:

STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is correct explanation of STATEMENT-1

Solution:

$$NO_2 \xrightarrow{\text{slow}} NO + O$$
$$CO + O \xrightarrow{\text{fast}} CO_2$$

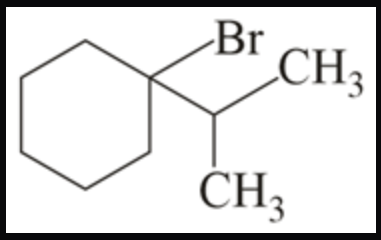
Q.40 Answer:

Neutral oxides CO_2, CO, BeO

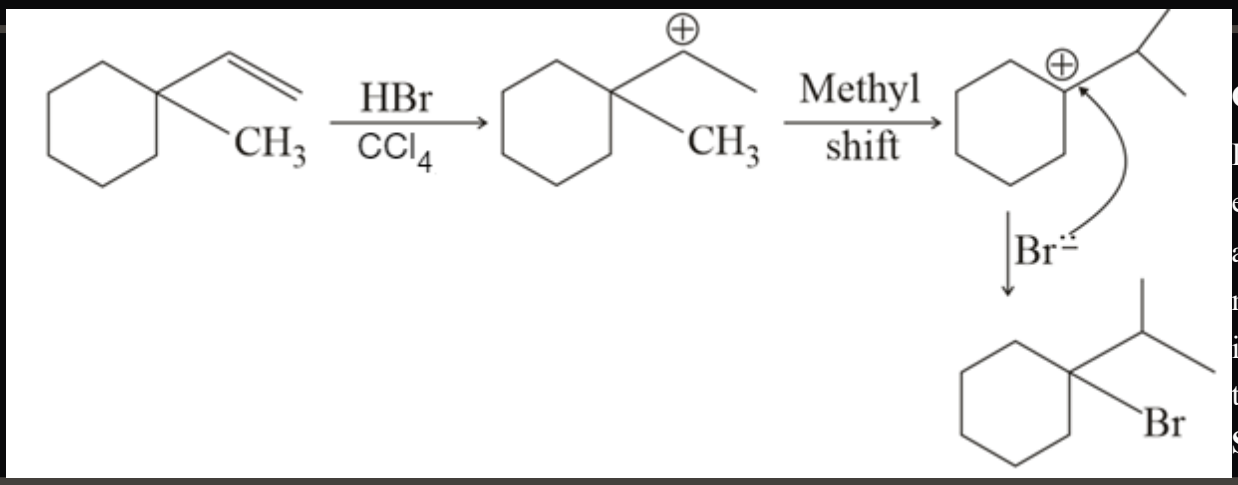
Solution:

CO_2 is an acidic oxide, CO is neutral and BeO is an amphoteric oxide.

Q.41 Answer:



Solution:



Q.42 Answer:

In a electrolysis experiment, α -amino acids migrate at the isoelectric point towards electrodes

Solution:

A. The pH at which Anion = Cation is called isoelectric point. At isoelectric point, α -amino acids do not migrate when electric field is applied.

B. $-SO_3H$ is strongly acidic and donates H^+ to weakly basic arylamino group. ArCOOH is not acidic enough to transfer H^+ to the arylamino group.

C. In $p-H_3NC_6H_4SO_3$, H_3N is acid enough to transfer H^+ to bases to give the soluble anion, $p-H_2N-C_6H_4SO_3\cdot -SO_3$ is too feebly basic and cannot accept H^+ from acids.

D. $\alpha-H_3N$ group increases acidity, because of its electron-withdrawing inductive effect.

Q.43 Answer:

$0.4\ N_A$

Solution:

$6.4\ \text{g of}\ CaC_2 \equiv 0.1\ \text{mol of}\ CaC_2$

$0.1\ \text{mol}$ **contains** $0.1\ N_A$ **molecules of** CaC_2

$\because 1$ **molecules have** 4π **electron** $\therefore 0.1\ N_A$ **molecule will have**

$0.1\ N_A \times 4 = 0.4\ N_A$

Q.44 Answer:

$Cr_2O_3\ (H_2O)_n$

Solution:

$Cr(OH)_3$ **is not precipitated, but the hydrated oxide,** $Cr_2O_3\ (H_2O)_n$ **is precipitated.**

$Cr^{3+} + 3OH^- \rightarrow Cr(OH)_3 \rightarrow Cr_2O_3\ (H_2O)_n$

Q.45 Answer:

Dipole-dipole interaction

Solution:

HF **is a polar molecule due to the difference in the electronegativity of the constituent atoms and the presence of a dipole in the molecule. So, the intermolecular forces among the** HF **molecules will be dipole-dipole interaction.**

Q.46 Answer:

—

Solution:

Molar mass of solute $= \frac{K_f \times w \times 1000}{\Delta T_f \times W}$

Given : $K_f = 1.86, w = 1.25\ g,\ W = 20\ g,$

$\Delta T_f = 273 - 271.9 = 1.1\ K$

Therefore, molar mass of solute

$= \frac{1.86 \times 1.25 \times 1000}{1.1 \times 20} = 105.7$

Q.47 Answer:

—

Solution:

Number of spectral lines $= \frac{(n_2-n_1)(n_2-n_1+1)}{2}$
$= \frac{(7-2)(7-2+1)}{2} = 15$

Q.48 Answer:

—

Solution:

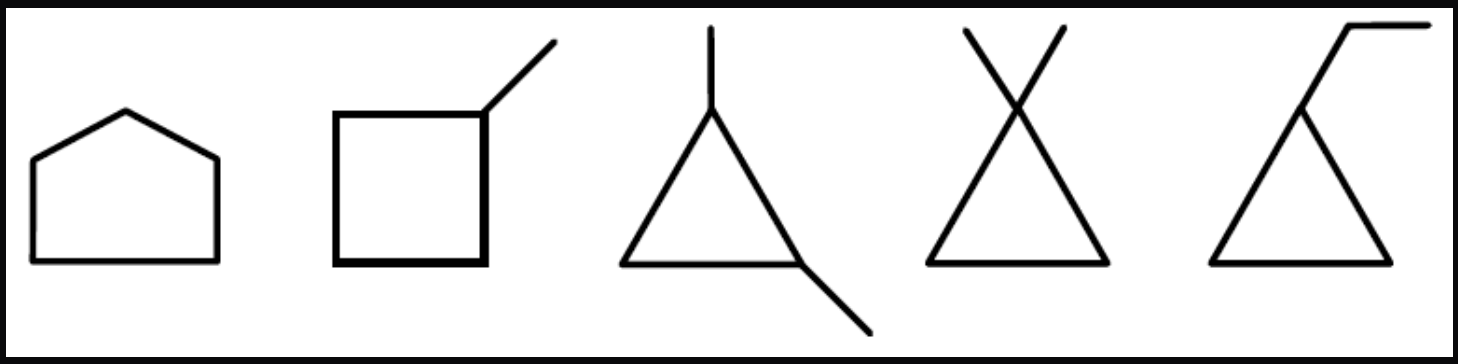
In general $t_{1/2}$ of reaction $\propto \frac{1}{(a_0)^{n-1}}$
For a second order reaction,
$t_{1/2} = \frac{1}{K(a_0)^{n-1}} = \frac{1}{K(a_0)} = \frac{1}{0.5\times0.2} = 10\,\text{min}$

Q.49 Answer:

—

Solution:

Cyclic C_5H_{10}



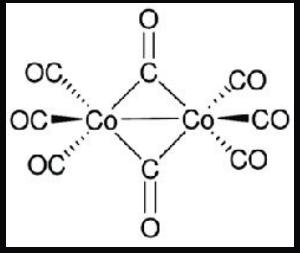
For 3rd structure 2 cis-trans and 1 optical isomer are possible. Total 7 isomers.

Q.50 Answer:

—

Solution:

Oxidation state of iron in $\left[Fe(CO)_2\left(NO^+\right)_2\right] = x + 2(0) + 2 = 0;\ x = -2$
So, EAN $= 26 - (-2) + 8 = 36 = x$



Oxidation state of cobalt in $Co_2(CO)_8 = 2x + 8(-0) =$ or $x = 0$
So, EAN $= 27 + 1 + 8 = 36 = y$
So, $x + y = 72$

Q.51 Answer:

1254

Solution:

The number of ways they can be arranged in a row is $\frac{9!}{2!3!4!} = 1260$

The three colour blocks of balls can be arranged in $3! = 6$ **ways.**

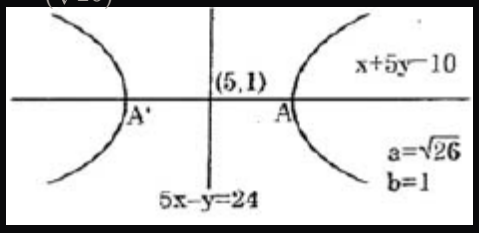
$\therefore$ **The desired number of ways is** $1260 - 6 = 1254.$

Q.52 Answer:

one vertex is $(0, 2)$

Solution:

$$\frac{\left(\frac{5x-y-24}{\sqrt{5^2+1^2}}\right)^2}{(\sqrt{26})^2} - \frac{\left(\frac{x+5y-10}{\sqrt{1^2+5^2}}\right)^2}{1} = 1$$



For vertex

$$\frac{x-5}{-\frac{5}{\sqrt{26}}} = \frac{y-1}{\frac{1}{\sqrt{26}}} = \pm\sqrt{26}$$

$$\Rightarrow (0, 2), (10, 0)$$

Q.53 Answer:

$-\cos x$

Solution:

$$f(x) = \cos x - \int_0^x (x-t)f(t)dt$$

$$\Rightarrow f(x) = \cos x - x\int_0^x f(t)dt + \int_0^x tf(t)dt$$

(differentiating w.r.t. x. by Leibnitz's Rule)

$$\Rightarrow f'(x) = -\sin x - xf(x) - \int_0^x f(t)dt + xf(x)$$

$$\Rightarrow f'(x) = -\sin x - \int_0^x f(t)dt$$

(again differentiating w.r.t.x)

$$\Rightarrow f''(x) = -\cos x - f(x)$$

$$\Rightarrow f''(x) + f(x) = -\cos x$$

Q.54 Answer:

-10

Solution:

Here, $f(x) = x^{3/2} + x^{-3/2} - 4\left(x + \frac{1}{x}\right)$

$f(x) = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^3 - 3\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) - 4\left(\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 - 2\right) \qquad [x > 0]$

Let $\sqrt{x} + \frac{1}{\sqrt{x}} = t,$

$\therefore \quad g(t) = t^3 - 3t - 4\left(t^2 - 2\right)$

Now,

$g(t) = t^3 - 4t^2 - 3t + 8$, where $t \in [2, \infty)$

$g'(t) = 3t^2 - 8t - 3 = (t - 3)(3t + 1)$

$g'(t) = 0 \Rightarrow t = 3 (t \neq -1/3)$

$g''(t) = 6t - 8$

$\Rightarrow \quad g''(3) = 10 > 0$

$\therefore$ Minimum value is

$g(3) = 27 - 36 - 9 + 8 = -10$

Hence, (a) is the correct answer.

Q.55 Answer:

5

Solution:

$\cos^{-1}\cos 12 + \sin^{-1}\sin 12 = 0$

$\therefore (k-2)x^2 + 8x + k + 4 > 0 \; \forall \; x \in R$

$\Rightarrow K - 2 > 0 \; \& \; 64 - 4(K-2)(K+4) < 0$

$\Rightarrow K > 4$

Q.56 Answer:

$\frac{181}{216}$

Solution:

Let the outcome on dice be n_1, n_2 and n_3. we first try to find the number of ways in which sum of outcomes is at the most equal to seven.

For this, $n_1 + n_2 + n_3 \leq 7$, where $n_i \in [1, 6]$.

$\Rightarrow (\mathrm{n}_1 - 1) + (\mathrm{n}_2 - 1) + (\mathrm{n}_3 - 1) + \mathrm{y}_4 = 4$

i.e., $y_1 + y_2 + y_3 + y_4 = 4,$

where y_1, y_2, y_3, y_4 are non-negative integers. This can happen in ${}^{4+4-1}\mathrm{C}_4$ i.e., ${}^{7}\mathrm{C}_4$ ways. Thus, required probability

$= 1 - \frac{{}^7C_4}{6^3} = \frac{181}{216}$

Q.57 Answer:

$3\bar{i} + \bar{j} - \bar{k}$

Solution:

$\vec{r} \times \vec{a} = \vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$

$\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

$\vec{r} \times (\vec{a} + \vec{b}) = 0$

$\vec{r} = k(\vec{a} + \vec{b})$

$\vec{r} \times \vec{a} = k[\vec{a} \times \vec{a} + \vec{b} \times \vec{a}] = \vec{b} \times \vec{a}$

$k(\vec{b} \times \vec{a}) = \vec{b} \times \vec{a}$

$k = 1$

$\vec{r} = \vec{a} + \vec{b} = 3\hat{i} + \hat{j} - \hat{k}$

Q.58 Answer:

Zero

Solution:

$\because a^{f(x)}$ **is always poisitive as** $a^x \in (0, \infty) \forall x \in R$

Minimum value of $g(x)$ **is** $= \frac{1}{2} \Rightarrow g(x)$ **is positive**

Since sum of two positive values is always positive, so it can never be zero

$\therefore$ **No solution.**

Q.59 Answer:

$\frac{1}{6}$

Solution:

$$\lim_{x \to 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4 \left(e^{2x^4} - 1 - 2x^4\right)} = \lim_{t \to 0} \frac{\sin t - t \cos t + t^5}{t \left(e^{2t} - 1 - 2t\right)} \quad \left\{\text{Let } x^4 = t\right\}$$

$$= \lim_{t \to 0} \frac{\left(t - \frac{t^3}{3!} + \frac{t^5}{5!} - \ldots..\right) - \left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \ldots..\right) + t^5}{t \left(1 + 2t + \frac{4t^2}{2!} + \frac{8t^3}{3!} + \frac{16t^4}{4!} + \ldots - 1 - 2t\right)} = \lim_{t \to 0} \frac{-\frac{t^3}{6} + \frac{t^3}{2} + \frac{t^5}{5!} - \frac{t^5}{4!} + \ldots + t^5}{2t^3 + \frac{8t^4}{3!} + \ldots}$$

$$= \frac{-\frac{1}{6} + \frac{1}{2}}{2} = \frac{-1 + 3}{12} = \frac{1}{6}$$

Q.60 Answer:

20

Solution:

Given,

Mean of three numbers $= 15$

Range $= 8$

Let the smallest number $= x$

We know that,

Range = Highest term − Lowest term

$\Rightarrow 8 =$ **Greatest number** $- x$

$\Rightarrow$ **Greatest number** $= x + 8$

According to question,

2nd number $- x = 1$

$\Rightarrow$ **2nd number** $= x + 1$

So, the three numbers are x, $x+1$ **and** $x + 8$.

Also, we know that,

Mean $= \frac{\text{Sum of all terms}}{\text{Number of terms}}$

$\Rightarrow 15 = \frac{\text{Sum of all terms}}{3}$

$\Rightarrow$**Sum of terms** $= 45$

$\Rightarrow x + (x + 1) + (x + 8) = 45$

$\Rightarrow 3x + 9 = 45$

$\Rightarrow 3x = 45 - 9$

$\Rightarrow x = \frac{36}{3}$

$\Rightarrow x = 12$

Hence, the greatest number is $= 12 + 8 = 20$

Q.61 Answer:

-9

Solution:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} = a \quad \ldots (i)$$

$$\frac{x-4}{1} = \frac{y-6}{2} = \frac{z+\lambda}{2} = b \quad \ldots (ii)$$

by (i) and (ii) : $\mathrm{a} = 2; \mathrm{b} = 1 \Rightarrow 4\mathrm{a} + 3 = 2\ \mathrm{b} - \lambda$

$\Rightarrow \lambda = -9$

Q.62 Answer:

there exists infinitely many B 's such that $AB = BA$

Solution:

$\because A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ **and** $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$.

Now

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

$$= \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix}$$

and

$$BA = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$
$$= \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}$$

If $AB = BA \Rightarrow a = b$

Hence $AB = BA$ is possible for infinitely many $B's$.

Q.63 Answer:

$\sqrt{x^2 + y^2} + y = cx^2$

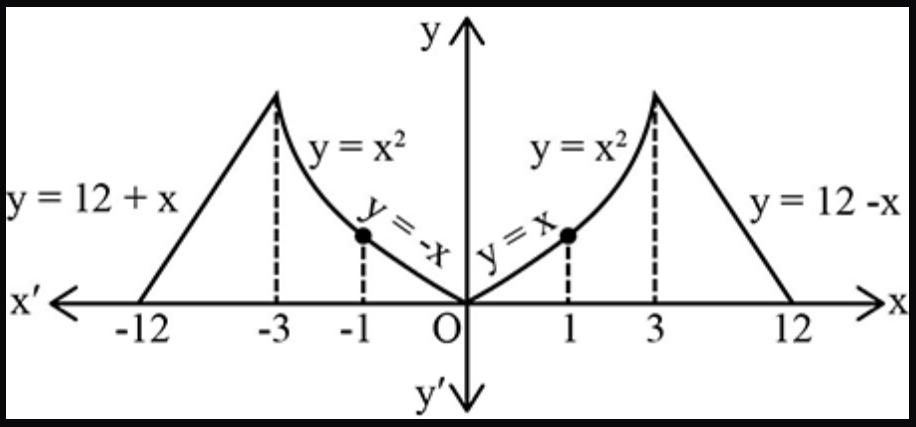
Solution:

$$\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}, y = vx \Rightarrow \frac{xdv}{dx} + v = v + \sqrt{1 + v^2}$$
$$\Rightarrow \int \frac{dv}{\sqrt{1 + v^2}} = \int \frac{dx}{x} \Rightarrow \ln\left(v + \sqrt{1 + v^2}\right) = \ln x + \ln c$$
$$\Rightarrow v + \sqrt{1 + v^2} = cx \Rightarrow y + \sqrt{x^2 + y^2} = cx^2$$

Q.64 Answer:

equal to $\{-3, -1, 0, 1, 3\}$

Solution:



Here, $f(x) = \begin{cases} 12 + x & -12 \leq x < -3 \\ x^2 & -3 \leq x \leq -1 \\ |x| & -1 < x < 1 \\ x^2 & \\ 12 - x & 3 < x \leq 12 \end{cases}$

Hence, $f(x)$ is not differentiable at $x = -3, -1, 0, 1, 3$

$\Rightarrow S = \{-3, -1, 0, 1, 3\}$

Q.65 Answer:

5

Solution:

We have $n(A_i) = i + 2$

$\therefore n(A_1) = 3, n(A_2) = 4, n(A_3) = 5$

similarly, $n(A_{100}) = 102$

given $A_1 \subset A_2 \subset A_3 \ldots\ldots\ldots\ldots A_{100}$

as A_3 is a subset of all the successive sets

$\therefore A_3 \cap A_4 \cap A_5 \cap \ldots\ldots\ldots \cap A_{100} = A_3$

$\therefore A_3 = A$

$\therefore n(A) = n(A_3) = 5$

Q.66 Answer:

25

Solution:

Given expansion: $\left(\frac{3}{2} + \frac{x}{3}\right)^n$

For numerically greatest term (T_{r+1}) in the expansion of

$(a + b)^n$

$r < \frac{n+1}{1+\left|\frac{a}{b}\right|} < r + 1$

$\Rightarrow 5 < \frac{n+1}{1+\left|\frac{\frac{3}{2}}{\frac{x}{3}}\right|} < 6$

Put $x = 3$,

$\Rightarrow 5 < \frac{2}{5}(n + 1) < 6$

$\Rightarrow \frac{25}{2} < n + 1 < 15$

$\Rightarrow \frac{23}{2} < n < 14$

$\therefore \ n = 12 \text{ or } 13$

So, the sum of possible values of

$n = 12 + 13 = 25$

Q.67 Answer:

$3\pi a^2$

Solution:

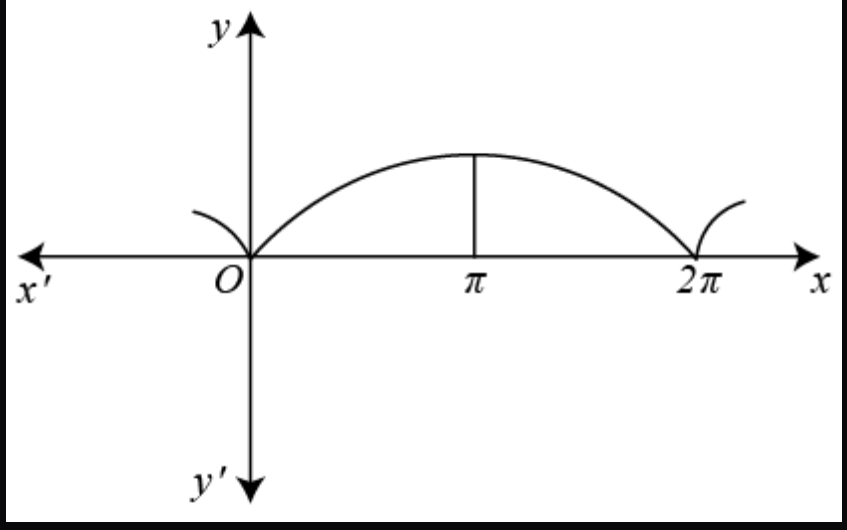
$$A = \int y dx$$

$$= a^2 \int_0^{2\pi} (1 - \cos\theta)^2 d\theta = 4a^2 \int_0^{2\pi} \sin^4 \frac{\theta}{2} d\theta$$

$$\theta = 2\phi \Rightarrow A = 8a^2 \int_0^{\pi} \sin^4 \phi d\phi$$

$$= 16a^2 \int_0^{\frac{\pi}{2}} \sin^4 \phi d\phi$$

$$= 16a^2 \times \frac{3.1}{4.2} \times \frac{\pi}{2} = 3\pi a^2.$$



Q.68 Answer:

1

Solution:

$$z = x + iy \Rightarrow \frac{iz+1}{iz-1} = \frac{1-y+ix}{-1-y+ix} = \frac{(1-y+ix)(ix+y+1)}{-(x^2+(y+1)^2)}$$

$$\text{Real part } = 2 \Rightarrow 1 - y^2 - x^2 = -2\left(x^2 + (y+1)^2\right)$$

$$\Rightarrow x^2 + y^2 + 4y + 3 = 0 \Rightarrow x^2 + (y+2)^2 = 1, \text{ a circle with radius 1.}$$

Q.69 Answer:

$\frac{4}{17}$

Solution:

A_1 : **Missing of spade** ; A_2 : **Missing card is not a spade**

E : 1st **card drawn is spade**

F : 2nd **card drawn is spade**

$$\mathrm{P}\left(\mathrm{B}_1\right) = \mathrm{P}\left(\frac{\mathrm{A}_1}{\mathrm{E}}\right) = \left[\frac{\frac{1}{4} \times \frac{12}{51}}{\frac{1}{4} \times \frac{12}{51} \times \frac{3}{4} \times \frac{13}{51}}\right] = \frac{12}{51} = \frac{4}{17}$$

$$\mathrm{P}\left(\mathrm{B}_2\right) = \mathrm{P}\left(\frac{\mathrm{A}_2}{\mathrm{E}}\right) = 1 - \mathrm{P}\left(\frac{\overline{\mathrm{A}}_2}{\mathrm{E}}\right) = 1 - \mathrm{P}\left(\frac{\mathrm{A}_1}{\mathrm{E}}\right) = \frac{13}{17}$$

$$\mathrm{P}(\mathrm{F}) = \mathrm{P}\left(\mathrm{B}_1\right) \cdot \mathrm{P}\left(\frac{\mathrm{F}}{\mathrm{B}_1}\right) + \mathrm{P}\left(\mathrm{B}_2\right) \cdot \mathrm{P}\left(\frac{\mathrm{F}}{\mathrm{B}_2}\right)$$

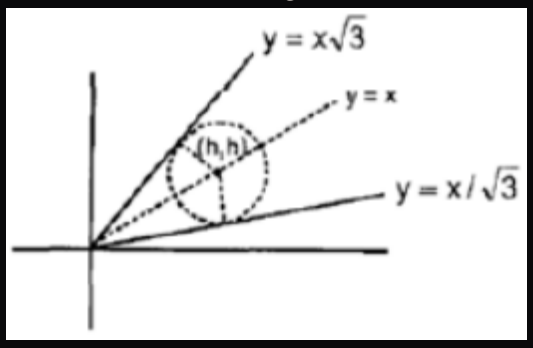
$$= \frac{4}{17} \times \frac{11}{50} + \frac{13}{17} \times \frac{12}{50} = \frac{4 \times 50}{17 \times 50} = \frac{4}{17}$$

Q.70 Answer:

$x^2 + y^2 - 2x(\sqrt{3}+1) - 2y(\sqrt{3}+1) + 7 + 4\sqrt{3} = 0$

Solution:

$\because$ centre lies on line $y = x$



$\therefore$ let centre (h, h)

$\therefore \frac{|n-n\sqrt{3}|}{2} = 1$

$\Rightarrow \mathrm{h} = (\sqrt{3}+1)$

$\therefore$ equation of required circle is

$x^2 + y^2 - 2x(\sqrt{3}+1) - 2y(\sqrt{3}+1) + 7 + 4\sqrt{3} = 0$

Q.71 Answer:

—

Solution:

$f(xf(y)) = x^p y^4,$ put $x = \dfrac{1}{f(y)}$

$\therefore \quad f(1) = \left(\dfrac{1}{f(y)}\right)^p y^4 = \dfrac{y^4}{(f(y))^p}$

For $\quad y = 1, \quad f(1) = \dfrac{1}{(f(1))^p} \Rightarrow f(1) = 1$

$\therefore \quad f(y) = y^{4/p} \ldots (1)$

$\therefore \quad f\left(xy^{4/p}\right) = x^p y^4$

Put $\quad y = z^{p/4}$

$\therefore \quad f(xz) = x^p z^p \Rightarrow f(x) = x^p \ldots (2)$

$\dfrac{4}{p} = p \quad \Rightarrow \quad p = 2$

Q.72 Answer:

—

Solution:

$f(x+y) = f(x) + 3y^2 + kxy$

In (i) **,put** $x = 1, f(1+y) = f(1) + 3y^2 + ky$

$\Rightarrow f(y+1) = 3y^2 + ky + 1 \ldots\ldots (ii)$

In (ii) **,put** $y = 1, f(2) = 3(1) + k + 1 = 6$

$\Rightarrow k = 2$

Using (ii) , $f(y+1) = 3y^2 + 2y + 1$

Replace $y \to x - 1$

$\Rightarrow f(x) = 3(x-1)^2 + 2(x-1) + 1 = 3(x-1)^2 + 2x - 1$

Now $\int_{-1}^{2} f(x)dx = \int_{-1}^{2} \left(3(x-1)^2 + 2x - 1\right) dx$

$= \left[(x-1)^3 + x^2 - x\right]_{-1}^{2} = [(8+1) + (4-1) - (2+1)] = 9$

Q.73 Answer:

—

Solution:

Given, $\begin{bmatrix} 1 & 25 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}^5 \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}^{10} \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}^5 \begin{bmatrix} x^2 - 5x + 20 \\ x + 2 \end{bmatrix} = [40]$

Let $B = \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$ and $C = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$

Here, $\mathrm{BC} = \mathrm{CB} = \mathrm{I}$

$\therefore B^5 C^{10} B^5 = I^{10} = I$

So, we get $\begin{bmatrix} 1 & 25 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x^2 - 5x + 20 \\ x + 2 \end{bmatrix} = \begin{bmatrix} 40 \end{bmatrix} \Rightarrow$

$\begin{bmatrix} 1 & 25 \end{bmatrix} \begin{bmatrix} x^2 - 5x + 20 \\ x + 2 \end{bmatrix} = [40]$

$\Rightarrow 1\left(x^2 - 5x + 20\right) + 25(x+2) = 40 \Rightarrow x^2 + 20x + 30 = 0$

$\therefore (1-\alpha)(1-\beta) = 1 - (\alpha + \beta) + \alpha\beta = 1 - (-20) + 30 = 51$

Q.74 Answer:

—

Solution:

$f(x) = x^{2013} + e^{2014x}$

$f'(x) = 2013x^{2012} + 2014e^{2014x} > 0$

$\Rightarrow$ $f(x)$ **is increasing function.**

Q.75 Answer:

—

Solution:

$a_1, a_2, a_3 \ldots . .. a_{49}$ **be in A.P.**

Let C.D be d

$\text{Now } \sum_{k=0}^{12} a_{4k+1} = 416$

$a_1 + a_5 + a_9 \ldots . + a_{49} = 416$

$\text{C.D } = 4\,d \quad n = 13$

$\frac{13}{2}[2a_1 + (12)4d] = 416$

$a_1 + 24d = 32 \quad \ldots (1)$

Also $a_9 + a_{43} = 66$

$a_1 + 8d + a_1 + 42d = 66$

$a_1 + 25d = 33 \quad \ldots (2)$

solving (1) and (2)

$a_1 = 8, d = 1$

Now :

$a_1^2 + a_2^2 + \ldots a_{17}^2$

$\Rightarrow 8^2 + 9^2 + \ldots . + 24^2$

$\Rightarrow \left(1^2 + 2^2 \ldots . + 24^2\right) - \left(1^2 + 2^2 \ldots . + 7^2\right)$

$\frac{24(25)(49)}{6} _ \frac{7(8)(15)}{6} = 4760$

$\Rightarrow 140\,m = 4760$

$m = 34$